PREDICTIVE MODELLING

DSBA

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DATE: 10/07/2025

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1 PROBLEM STATEMENT

Analysis of Driver Variables for First-Day Content Viewership on ShowTime OTT Platform using linear regression model.

1.1 Context

Over-the-top (OTT) services use the internet to offer digital television content directly to viewers, bypassing traditional cable and satellite providers. With the increasing desire for flexible, on-demand entertainment, OTT services have seen fast global adoption. Viewers may now enjoy original and licensed content on a variety of platforms, including smartphones, Smart TVs, and desktop computers. Changing media consumption habits, aided by events like as the COVID-19 epidemic, have increased OTT usage. Many outlets reported increased viewership and subscriptions, indicating a clear transition from traditional broadcasting to digital streaming. As the industry evolves, knowing the drivers of content engagement, especially elements driving day-one viewership and it is critical for platform growth and strategic planning.

1.2 Objective

The purpose of the analysis is to identify and evaluate the key factors driving first-day viewership of content on ShowTime, an OTT service provider. This study will employ a linear regression model to analyse the impact of variables such as user platform engagement, advertising strategies, content scheduling, and external factors such as weekends and vacations. The model's findings will help ShowTime design focused strategies to boost content visibility and overall platform engagement, resulting in higher first-day viewership.

1.3 Data Dictionary

The dataset covers the various distinct factors for content analysis. The detailed data dictionary is provided below:

1. **visitors:** Average number of visitors, in millions, to the platform in the past week
2. **ad_impressions:** Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
3. **major_sports_event:** Any major sports event on the day
4. **genre:** Genre of the content
5. **dayofweek**: Day of the release of the content
6. **season:** Season of the release of the content
7. **views_trailer:** Number of views, in millions, of the content trailer
8. **views_content:** Number of first-day views, in millions, of the content

1.4 Data Structure

We will upload the dataset and check it by using head() and tail() to see first and last 5 rows, we will also check data types using info() and shape attribute in pandas to identify the dataset total numbers or rows and columns.
Top 5 rows :

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

Table 1. Top 5 rows.

Bottom 5 rows:

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
995	1.58	1311.96	0	Romance	Friday	Fall	48.58	0.36
996	1.34	1329.48	0	Action	Friday	Summer	72.42	0.56
997	1.62	1359.80	1	Sci-Fi	Wednesday	Fall	150.44	0.66
998	2.06	1698.35	0	Romance	Monday	Summer	48.72	0.47
999	1.36	1140.23	0	Comedy	Saturday	Summer	52.94	0.49

Table 2. Bottom 5 rows.

Shape:

Shape of dataset shows **1000 (rows) samples with 8 columns**.

Data Types:

There are 5 numerical columns (**visitors, ad_impressions, major_sports_event, views_trailer** and **views_content)** and 3 categorical columns (**genre**, **dayofweek**, **season**).

Statistical Summary:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
visitors	1000.0	NaN	NaN	NaN	1.70429	0.231973	1.25	1.55	1.7	1.83	2.34
ad_impressions	1000.0	NaN	NaN	NaN	1434.71229	289.534834	1010.87	1210.33	1383.58	1623.67	2424.2
major_sports_event	1000.0	NaN	NaN	NaN	0.4	0.490143	0.0	0.0	0.0	1.0	1.0
genre	1000	8	Others	255	NaN	NaN	NaN	NaN	NaN	NaN	NaN
dayofweek	1000	7	Friday	369	NaN	NaN	NaN	NaN	NaN	NaN	NaN
season	1000	4	Winter	257	NaN	NaN	NaN	NaN	NaN	NaN	NaN
views_trailer	1000.0	NaN	NaN	NaN	66.91559	35.00108	30.08	50.9475	53.96	57.755	199.92
views_content	1000.0	NaN	NaN	NaN	0.4734	0.105914	0.22	0.4	0.45	0.52	0.89

Table 3. Statistical Summary.

Observations of Statistical Summary

1. Visitors

Stats:

Mean = 1.70M, Median = 1.70M, Standard Deviation = 0.23M, Min = 1.25M, Max = 2.34M, IQR = 1.55M - 1.33M

Interpretation:

The data shows mean ≈ median (symmetric distribution) but has a low variance (SD = 0.23M) implies stable daily visitor volume. This tight IQR (a range of 0.28M) also verifies 50% of days had visitors between 1.55M - 1.33M, with rare extremes beyond this range.

2. ad_impressions

Statistics:

Mean = 1434.7, Median = 1383.6, Standard Deviation = 289.5, Min = 1010.9, Max = 2424.2, IQR = 1210.3 - 1623.7

Interpretation:

Data shows right-skewed distribution (mean > median) and high variability (SD = 289.5) suggest frequent fluctuations. The wide IQR (413.4 range) and max value (2,424.2, 49% above median) indicate occasional high-impact days, likely driven by external factors like vacations or events.

3. major_sports_event

Statistics:

Binary data (0/1) represents yes or no, Mean = 0.4, standard deviation = 0.49, Frequency = 40% (400 days).

Interpretation:

The dataset was dominated by non-event days (60%) with events occurring on 40% of days. In other metrics (such ad_impressions), this imbalance might aid in the analysis of event-driven traffic spikes.

4. Views Trailer

Statistics:

Mean = 66.91, Median = 53.96, Standard Deviation = 35.0, Min = 30.08, Max = 199.92, IQR = 50.94-57.75

Interpretation:

High standard deviation and extreme right-skew (mean >> median) indicate irregular views on trailer. Although 75% of days had ≤57.8 views, unusual outliers (max = 199.9, 246% above median) could indicate viral traction or marketing abnormalities.

5. Views Content

Statistics:

Mean = 0.47, Median = 0.45, Standard Deviation = 0.11, Min = 0.22, Max = 0.89, IQR = 0.40-0.52

Interpretation: Near-symmetric distribution (mean ≈ median) and low std dev denote consistent user engagement. The compact IQR (0.12 range) shows 50% of days had content views between 0.40-0.52, with no extreme outliers.

Categorical Columns Unique Values and counts:

```
Value counts for each categorical column:
genre         8
dayofweek     7
season        4
dtype: int64

 dayofweek
Friday        369
Wednesday     332
Thursday       97
Saturday       88
Sunday         67
Monday         24
Tuesday        23
Name: count, dtype: int64

 season
Winter    257
Fall      252
Spring    247
Summer    244
Name: count, dtype: int64

 genre
Others      255
Comedy      114
Thriller    113
Drama       109
Romance     105
Sci-Fi      102
Horror      101
Action      101
Name: count, dtype: int64
```

Table 4. Unique Values and Counts.

Observations on content release counts based on attributes categories

1. Dayofweek

Observations:

- Releases on Friday (36.9%) and Wednesday (33.2%) account for 70.1% of total material.
- Releases on Mondays (2.4%) and Tuesdays (2.3%) are infrequent.
- The weekend (Saturday: 8.8%, Sunday: 6.7%) and midweek (Thursday: 9.7%) are secondary options.

Inferences:

- In order to reach as many people as possible, publishers should strategically time their releases to coincide with midweek (Wednesday) and weekend transitions (Friday).
- Avoiding early-week releases: Monday and Tuesday appear to be low-engagement days, as evidenced by the few releases on these days.
- Bias alert: Wed/Fri patterns will be over-represented in the "release day impact" analysis.

2. season

Observations:

- Winter (25.7%), Fall (25.2%), Spring (24.7%), and Summer (24.4%) are in almost perfect balance.
- The maximum and minimum differences between winter and summer are 1.3%.

Inferences:

- Seasons have equal distributions of content, suggesting so seasonal preferences.

3. genre

Observations:

- "Others" is the most common (25.5%): This genre category is 2.2 times larger than normal.
- There is balance among other genres:
- Drama (10.9%), Thriller (11.3%), and Comedy (11.4%)
- Horror (10.1%), Action (10.1%), Sci-Fi (10.2%), and Romance (10.5%).
- The genre range (excluding "Others") is 10.1-11.4% (a difference of 1.3%).

Inference:

- The content release is evenly distributed across major genres, but 1 in 4 releases is uncategorized ("Others").

2 EXPLORATORY DATA ANALYSIS

2.1 Univariate Analysis

Distribution of visitors:

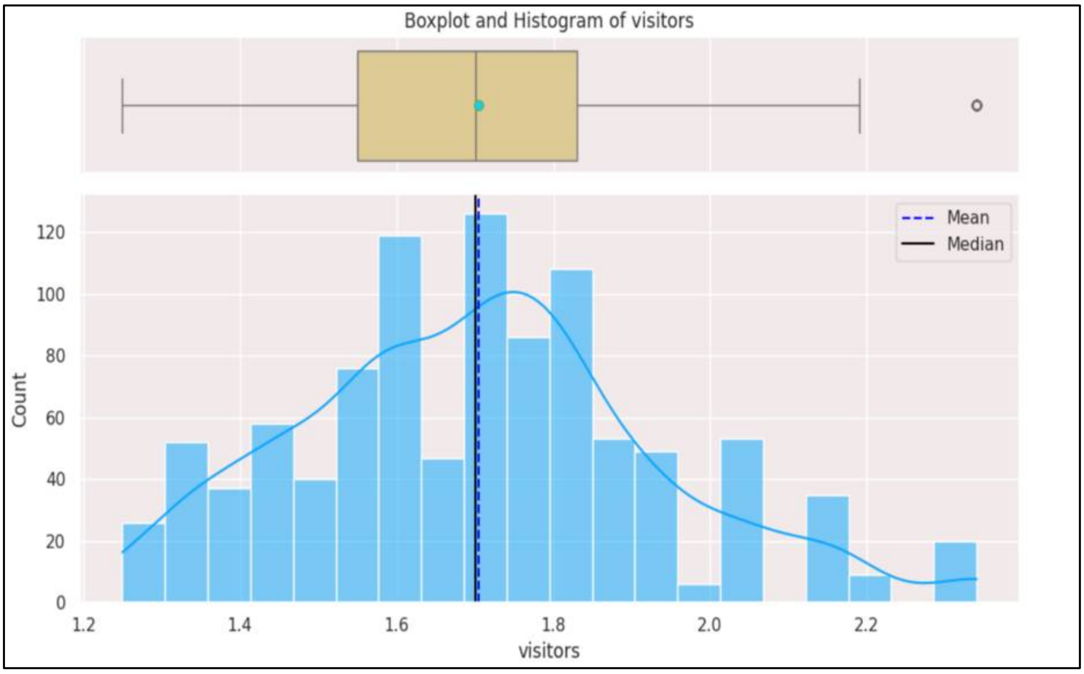

Figure 1. Distribution of visitors.

Observations

Histogram plot:

- The histogram depicts the distribution of visitors ranging from 1.2 to 2.2 million.
- The distribution appears to be slightly right-skewed, with a greater concentration of values between 1.5 and 1.8 million visitors.
- The majority of visitors range from 1.6 to 1.8 million, with a peak of about 1.7 million.

- There are fewer examples of content with either low or extremely high visitor counts (less than 1.3 million or more than 2.0 million).

Boxplot:

- The boxplot verifies that the distribution of visitor counts is slightly right skewed, as indicated by the histogram. The median is slightly to the left of the box, indicating the presence of a tail on the right, or upper end of the distribution.
- Median: The median visitor count seems to be approximately 1.7 million.
- Mean (gold dot) is equal to median.
- Outliers: There appears to be some outliers present on the higher end (beyond upper whiskers) of the visitor counts. This implies that there may be a few content releases that received much more visitors than the average content.

Distribution of ad_impressions:

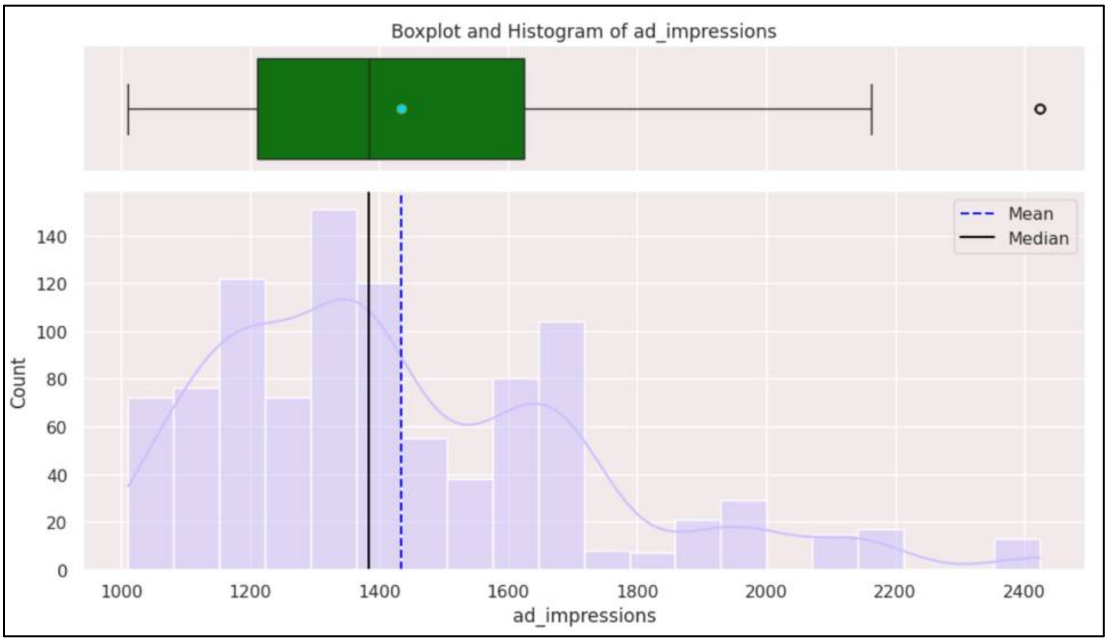

Figure 2. Distribution of ad_impressions.

Observations

Histogram:

- The Plot depicts data is right-skewed, with the majority of values ranging between 1000 and 1500. Peaks around 1200, 1400 and 1600.
- The long tail stretches right till 2400, indicating fewer high-impression campaigns.

Boxplot:

- Distribution is right skewed with median around 1400. Mean (gold dot on boxplot) > median.
- There is an outlier present beyond 2400, above upper whiskers, suggesting higher ad impression for certain campaigns.

Distribution of major_sports_event:

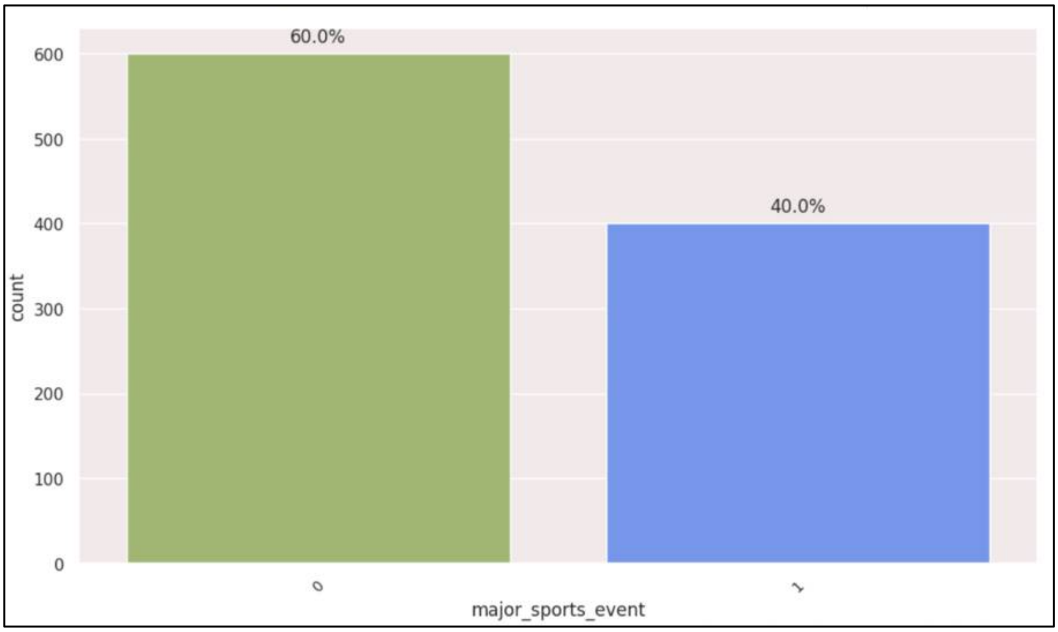

Figure 3. Distribution of major_sports_event.

Observations

- The bar plot shows there would be 60% content release on normal day, while 40% of releases will be on major sports event.
- **Note:** major_sports_event day: 1 = Yes, 0 =No.

Distribution of dayofweek:

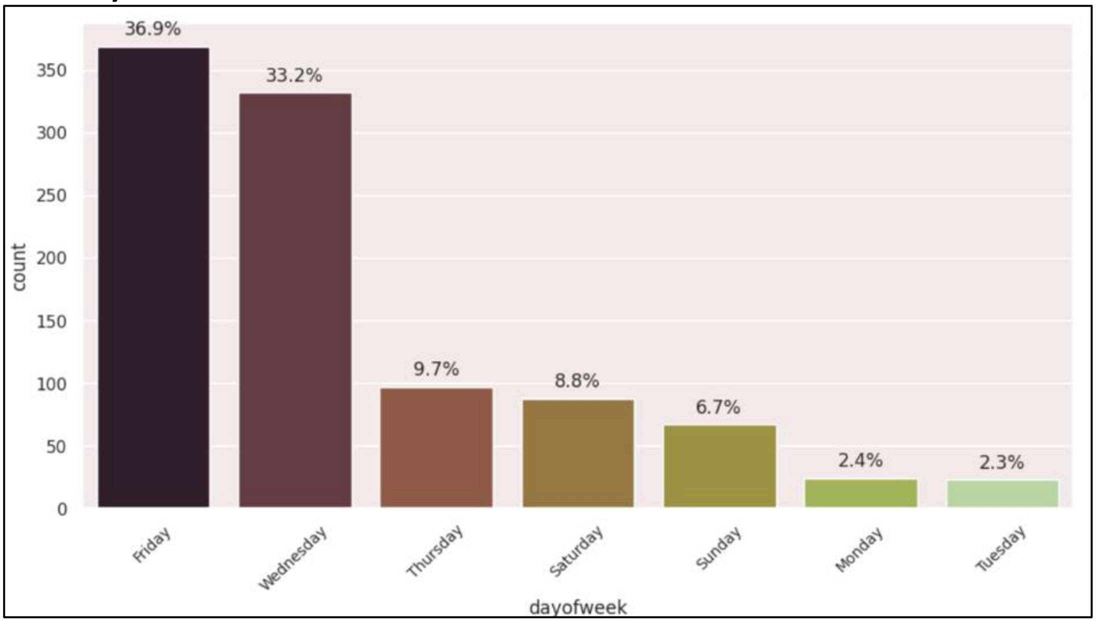

Figure 4. Distribution of dayofweek.

Observations

- Day of the Week is a nominal categorical variable having a bimodal, unbalanced distribution.

- The majority of content is released on Fridays (37%), followed by Wednesdays (33%), with Mondays and Tuesdays being the least used, showing that Friday and Wednesday are prime viewing days.

Distribution of season:

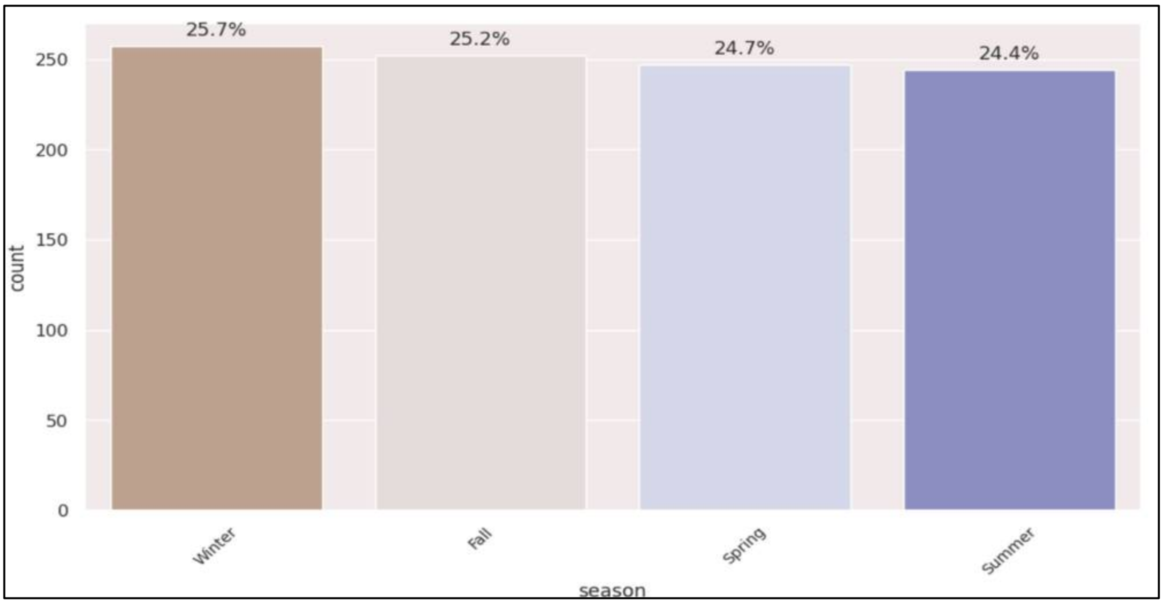

Figure 5. Distribution of seasons.

Observations

- Winter (25.7%), Fall (25.2%), Spring (24.7%), and Summer (24.4%) releases were essentially flat, indicating that the factor is well balanced for modelling.
- Plot shows nearly uniform distribution.
- High content release during winter and summer seasons.
- Autumn and spring are the seasons with the least content release.
- This could be due to a seasonal shift in the viewer's content preferences.

Distribution of view_trailer:

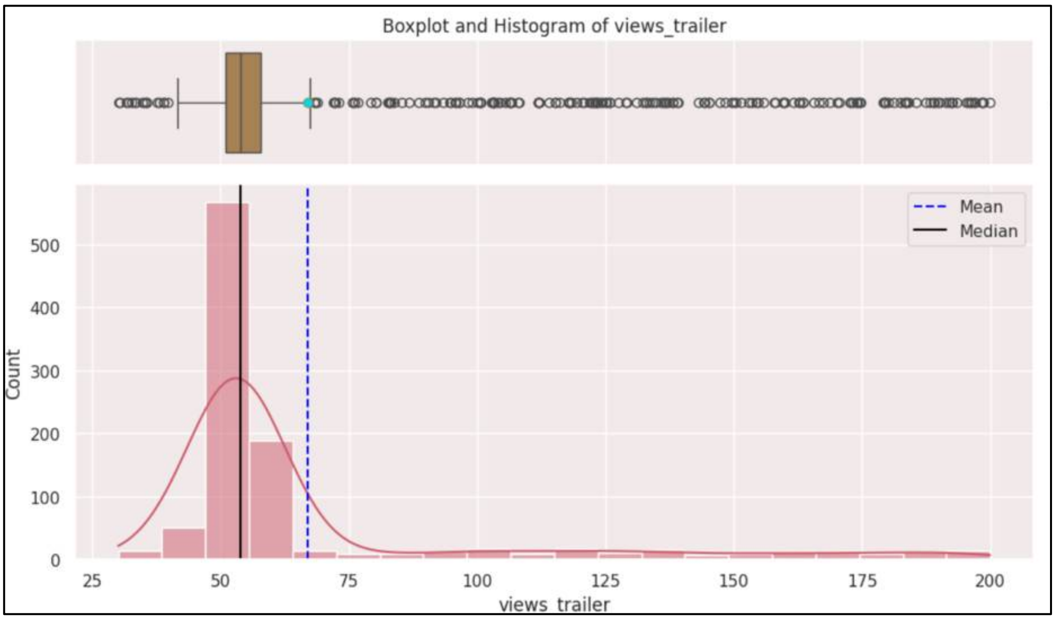

Figure 6. Distribution of view_trailer.

Observations

Histogram:

- The distribution of trailer views is highly right-skewed, with the majority of the data concentrated between 50 million and 75 million views, indicating that most content receives less trailer views, while a handful receive much more.
- The median trailer views are approximately 53.96 million, with a mean of 66.91 million.

Boxplot:

- The boxplot reveals a considerable number of outliers that exceed 75 million views, with a maximum value of 199.92 million.
- These outliers have much greater trailer views, possibly due to popularity or extensive marketing efforts.

Distribution of views_content:

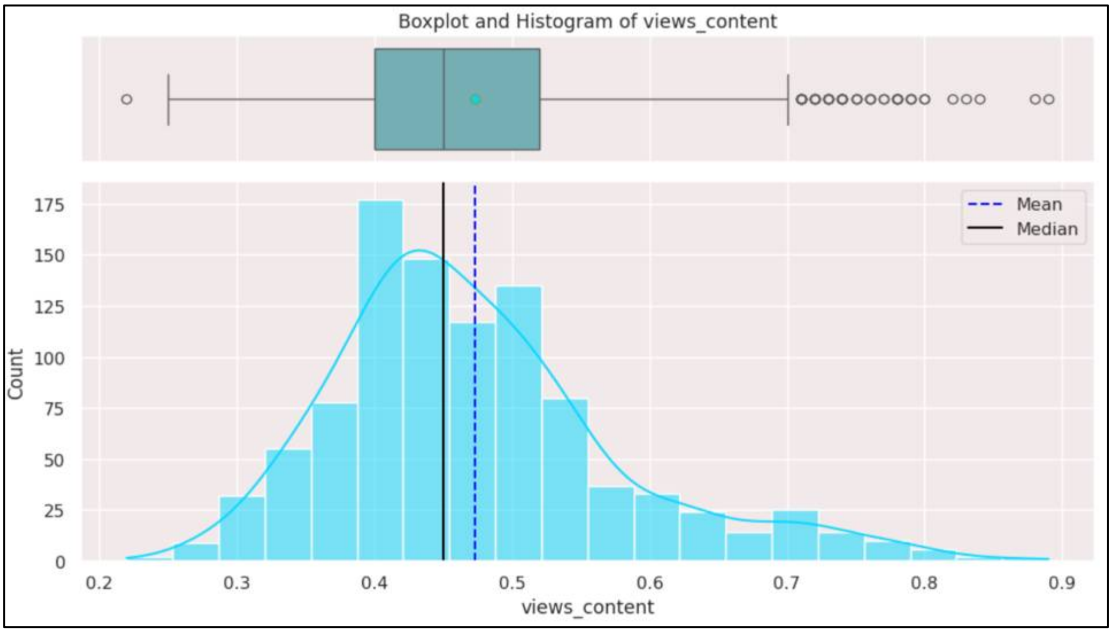

Figure 7. Distribution of views_content.

Observations

Histogram:

- The distribution of first day content view has moderate right-skewness and approximately normal, with majority of content views is concentrated between 0.4M to 0.5M.
- First-day content views with Mean = 0.47 and Median = 0.45M.

Boxplot:

- The boxplot reveals several outliers on the beyond upper whiskers with 0.7-0.9 million view on first-day of content release and one outlier outside the lower whiskers with approximately 0.2 million views on first-day of content release, potentially due to popularity, strong marketing, or other factors.

- Overall, the content views data is relatively normally distributed with a slight right skew, indicating a typical range of content views values around the mean.
- Analysing the relationship between trailer views and first-day content views can reveal whether good trailer promotion influences overall viewership.

Distribution of genre:

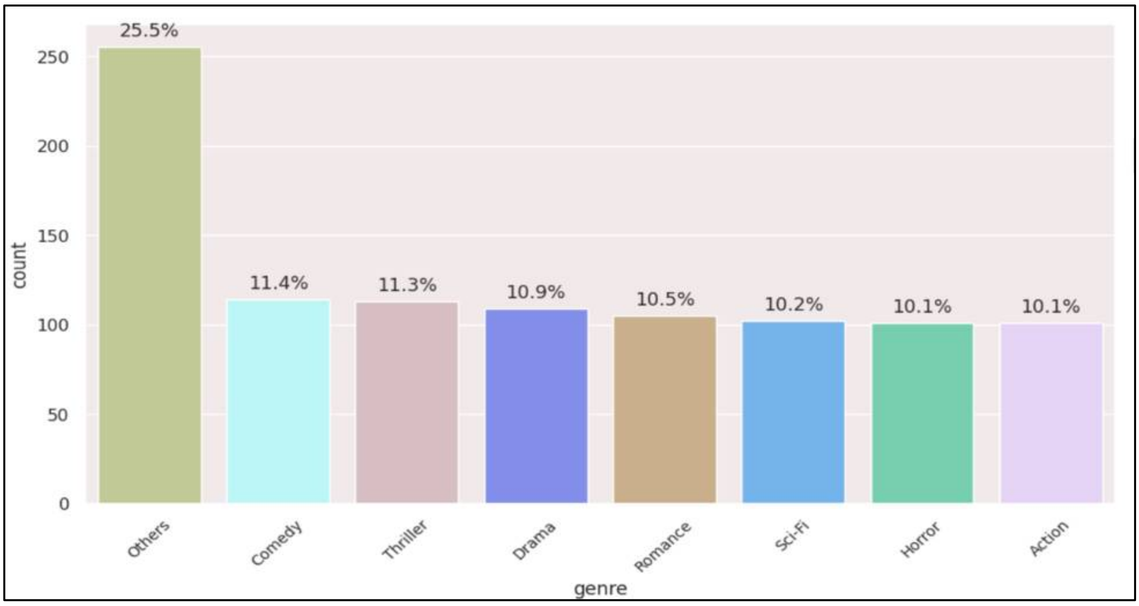

Figure 8. Distribution of genre.

Observations

- The bar chart represents a long-tail catalogue with unbalanced distribution.
- "Others" genre takes the lead with 26% of content release in this category, followed by Comedy (11.4%), Thriller (11.3%),Drama (10.9%) and Romance (10.5%).
- Others genre might represent significant content that doesn't fit traditional genres.
- The amount of content releases in the action, horror, and science fiction genres is relatively low (10.1-10.2%).
- Understanding the popularity of various genres might be useful in content strategy.

2.2 Bivariate Analysis

Heatmap Correlation between all numerical columns:

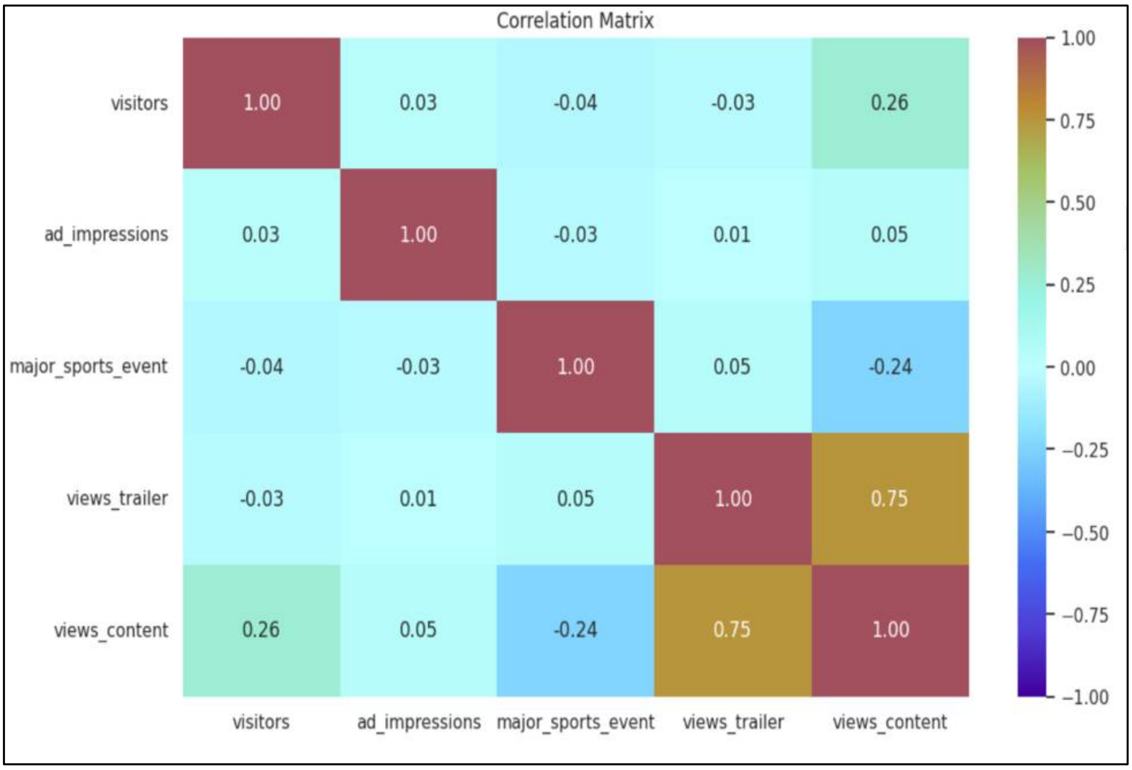

Figure 9. Heatmap Correlation.

Observations

views_trailer vs. views_content:

- views_trailer and views_content (target variable) show a strong positive linear correlation (r = +0.75), indicating that more views on trailer of the content result in more first-day views of the content.
- This means that as the number of trailer views increases, so do first-day content views. This shows that trailer popularity is the most important numeric predictor of content performance on Day One.

views_content vs. visitors:

- Views_content and visitors have weak positive correlation (r = +0.26).
- When platform traffic rises overall due to increases number of visitors, there is a minor trend for content to get more views.
- While the association is not particularly strong and significant, increased platform activity has a beneficial impact on day-one viewership.

ad_impressions vs. views_content:

- ad_impressions and views_content have a very weak positive correlation (r = +0.05).
- The number of ad impressions seems to have almost no linear relationship with first-day views.
- This shows that simply displaying ads is insufficient; their efficacy may be more dependent on content quality, targeting, or conversion to trailer views than mere volume.

major_sports_event vs. views_content:

- The chance of major sports event on the day of content release and the views on content has moderate negative correlation (r = -0.24).
- Launching content during a major sporting event results in significantly reduced day-one viewership.
- Content planners should avoid large match nights for flagship releases

major_sports_event vs. visitors, ad_impressions, views_trailer:

- major sport even on the day of release has very weak negative correlation with visitors (r = -0.04) and ad_impressions (r = -0.03), suggesting less traffic due to major sports event, and campaigns might tends to run marginally fewer impressions on sports day.
- major sport even on the day of release has very weak positive correlation with views_trailer (r = +0.05), indicates that trailers are still watched on sports events day but get less views than non-sports event days.

Inferences

- Target marketing to increase views_trailer, as trailer engagement is the primary numerical variable.
- Sports events degrade performance; avoid them for major content releases.
- There is no risk of multicollinearity since, with the exception of the deliberate 0.75 link to the target, all predictor-to-predictor correlations are close to zero, allowing all numerical features to remain safely in the regression model.

Distribution of visitors vs. views_content:

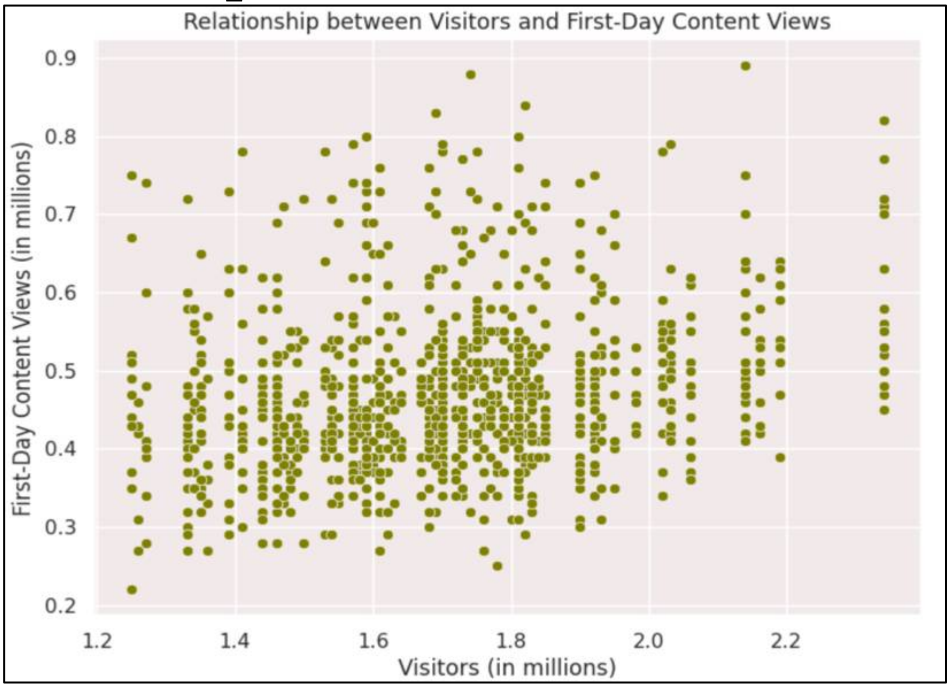

Figure 10. Distribution of visitors vs. views_content.

Observations

- Views_content and visitors have weak positive correlation (r = +0.26).
- Therefore, there no significant relationship between the visitors on the platform and first day content views.

Distribution of visitor vs. genre:

genre	count	mean	median	min	max	std
Action	101	1.740891	1.73	1.25	2.34	0.255946
Comedy	114	1.697456	1.70	1.26	2.34	0.254711
Drama	109	1.698440	1.70	1.25	2.34	0.210384
Horror	101	1.712376	1.73	1.26	2.34	0.216338
Others	255	1.690745	1.70	1.25	2.34	0.240437
Romance	105	1.716381	1.73	1.25	2.34	0.207513
Sci-Fi	102	1.707941	1.73	1.25	2.34	0.245451
Thriller	113	1.692920	1.69	1.27	2.34	0.209999

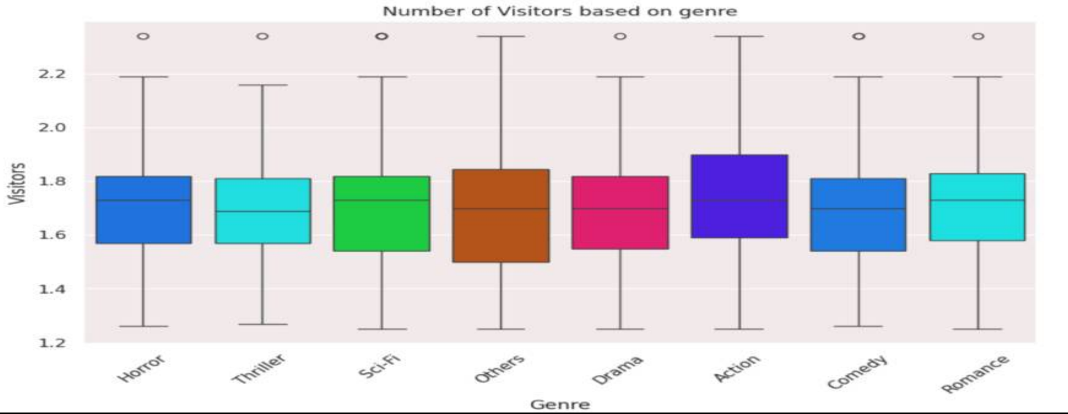

Figure 11. Distribution of visitor vs. genre.

Observations

- Action genre has the maximum visitors on an average (1.74 M).
- Other top performing genres on an average are Horror (1.71M), Sci-Fi (1.70M), Romance: 1.71M).
- Drama, Comedy, Others and Thriller genres as have relatively low visitors (approximately 1.69M) as compared to top genres attracting visitors.
- The box plot shows outliers for all genres having visitors above 2.2M except Others.
- Overall, clear viewer preference for Action, horror, Sci-fi and romance content.

Distribution of genre vs. views_content:

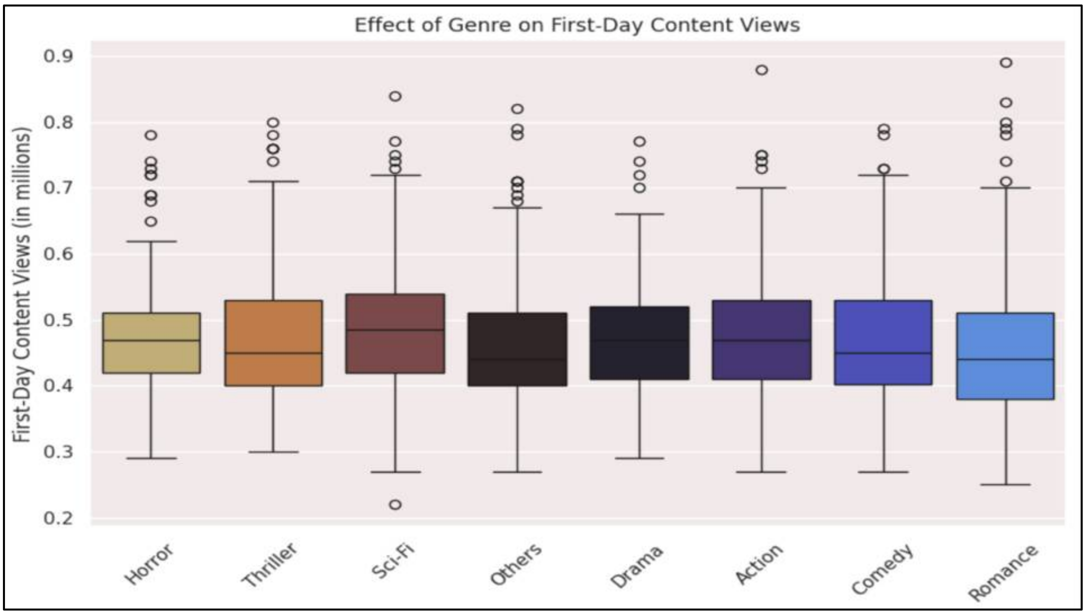

Figure 12. Distribution of genre vs. views_content.

Observations:

- The box plot shows that different genres have different impacts on first-day content views.
- Sci-Fi, Horror and Thriller are top performing genres in terms of first-day content viewership on an average (0.49M, 0.48M and 0.47M) even though their counts for content release is less compared to Other (count = 255).
- Others (0.45M) and Romance (0.46) has the lowest mean viewership.
- Understanding the popularity of different genres can help in content strategy.

Distribution of major_sports_event vs. views_content:

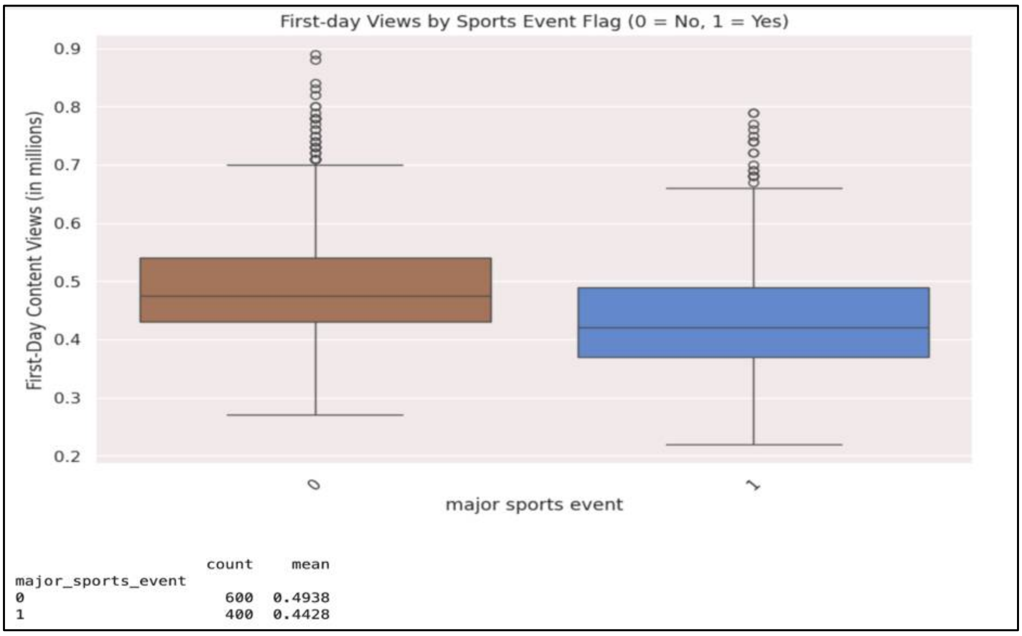

Figure 13. Distribution of major_sports_event vs. views_content.

Observations

- The mean First-day content views when no sports event is 0.49M, while for major sport event day view is 0.44M.
- There are several outliers on upper whiskers of both the box plots.
- Outliers on non-sports day shows viewership ranging between 0.7-0.9M, while for sport event day it ranges between 0.65M-0.8M.
- This suggests there is significant impact of sports even on the first-day as it results in reduced content viewership.

Distribution of dayofweek vs. genre:

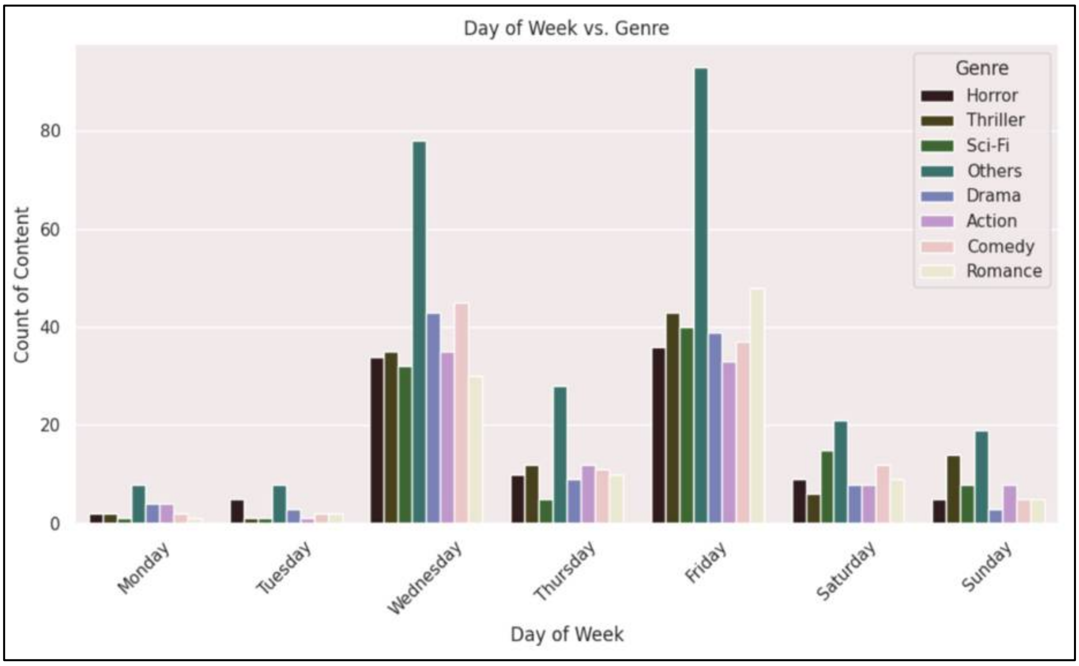

Figure 14. Distribution of dayofweek vs. genre.

Observations

- **Friday and Wednesday** have the biggest number of content releases in most genres.
- **Drama, Comedy, and Others** are the most popular genres on high-traffic days, such as Wednesday and Friday.
- **Action and Thriller** content is more evenly distributed, with modest surges on Friday and Wednesdays.
- **Horror and Sci-Fi** are more common on mid-to-late weekdays, whereas **Romance** has a balanced distribution but is the least popular.
- **Monday and Tuesday** saw the least content releases across all categories.

Interpretation

- The platform appears to prefer **Wednesday and Friday** for delivering high-volume content across different genres, most likely due to audience interaction patterns.
- **Weekends** have moderate genre activity (particularly Action and Sci-Fi), indicating deliberate placements for leisure-time content.

- Lighter genres, such as **Romance** and **Comedy**, are broadcast midweek, probably targeting casual weekday viewers.
- This analysis can inform **content scheduling**, **campaign targeting**, and **genre allocation** throughout the week.

Distribution of dayofweek vs. views_content:

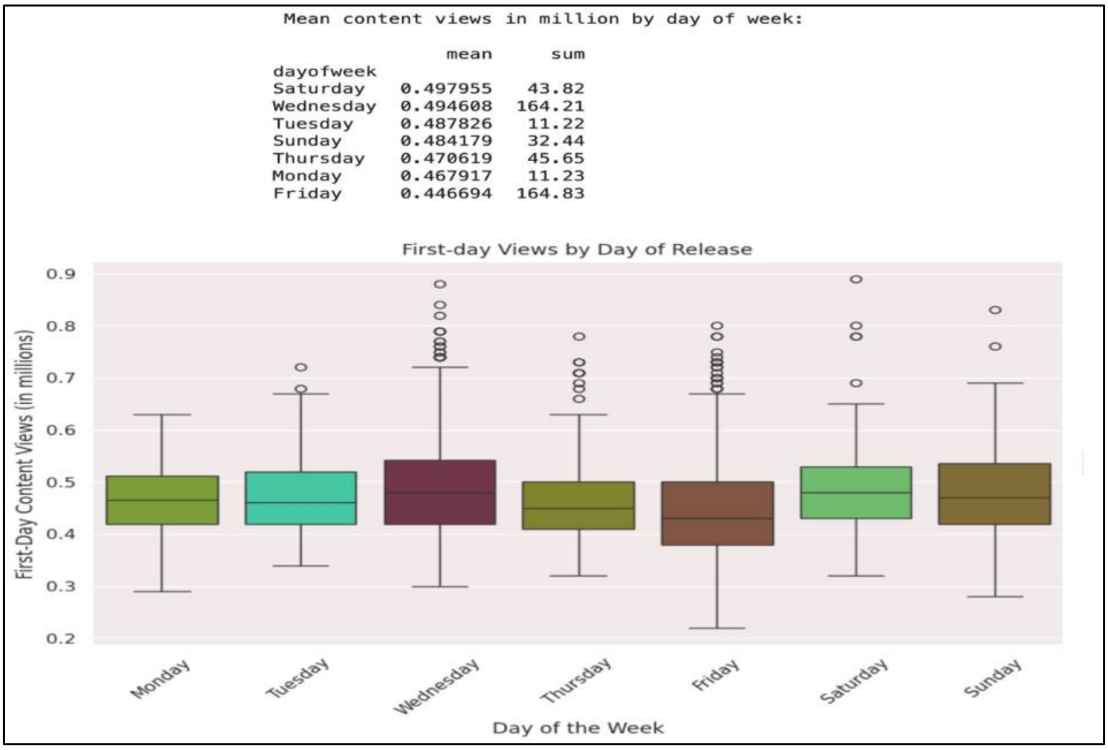

Figure 15. Distribution of dayofweek vs. views_content.

Observations

- In the boxplot, Saturday has the highest median and the highest average views per content (mean = 0.498M).
- Wednesday exhibits the most variety with several high-performing outliers and ranks closely in mean views (0.495M).
- Despite having the most views overall, Friday has the lowest average views per piece of content (0.447M), suggesting a large volume of releases.
- The fact that Monday and Tuesday have the fewest total views suggests that there may be fewer content drops on those days.
- Wednesday and Friday are the days with the most outliers, suggesting occasional viral content, leading to high first-say viewership.

Interpretation

- Viewership fluctuates dramatically each day of the week.
- While Friday has the most releases (total views), it underperforms per-content, indicating content saturation or audience fatigue.
- Saturday is the best day to optimise per-content performance, both in terms of average and boxplot distribution (high median and moderate spread).

- Wednesday is a good midweek candidate because of its high average and frequent spikes, which are probably caused by targeted content.
- Releasing significant or high-ticket content on Saturday or Wednesday may increase engagement.
- Friday may be set aside for larger, lower-priority releases in which performance is less crucial.

Distribution of season vs. views_content:

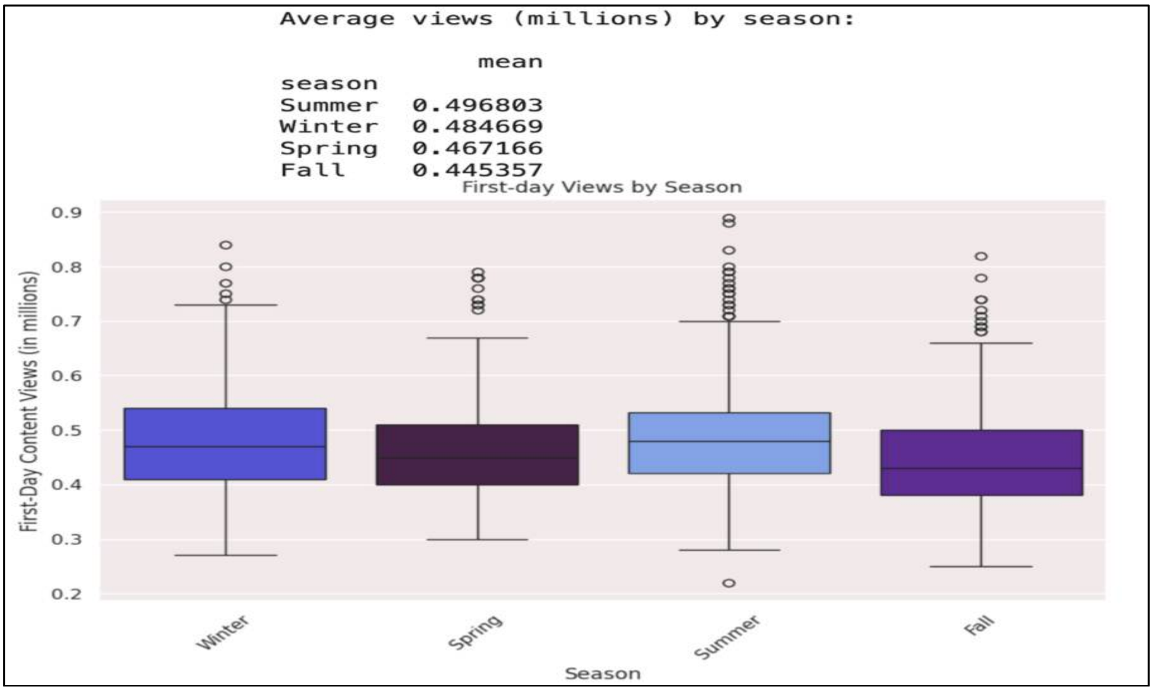

Figure 16. Distribution of season vs. views_content.

Observations

Summer

- Highest median viewership (0.45M) and mean viewership (0.497M).
- Outliers in both ends of whiskers indicate breakout potential (up to 0.7M+).

Winter

- Most consistent performance (narrow IQR). 2.3% below Summer median but reliable with 2nd highest mean viewership (0.485M).
- There are outliers on the upper whiskers in the range of 0.73M-0.85M.

Spring

- Spring ranks third in terms of mean (0.467M), with an IQR similar to Winter but a lower median.
- Outliers are present in upper whiskers of viewership range ~(0.71M-0.8M).

Fall

- Fall has the lowest mean (0.445M) and median, with a slightly narrower range that centres lower than other seasons.
- Outliers on upper end with viewership in range ~(0.65M-0.82M).

Summer and **Winter** have high outliers (≥ 0.85 million), indicating infrequent blockbuster releases.

Interpretation Seasonality affects average first-day views, which vary by around 11% between peak (Summer) and low (Fall) seasons.

Summer advantage: Increased leisure time (school breaks, vacations) is likely to improve engagement, making summer the best period for premium releases.

Winter opportunity: Winter's mean is similar to summer's, implying that vacation breaks and indoor entertainment fuel increased viewership.

Lower averages may indicate conflict with holiday preparations or new TV seasons. Reserve Autumn for lower-priority programs or allocate extra marketing.

Distribution of views_trailer vs. views_content:

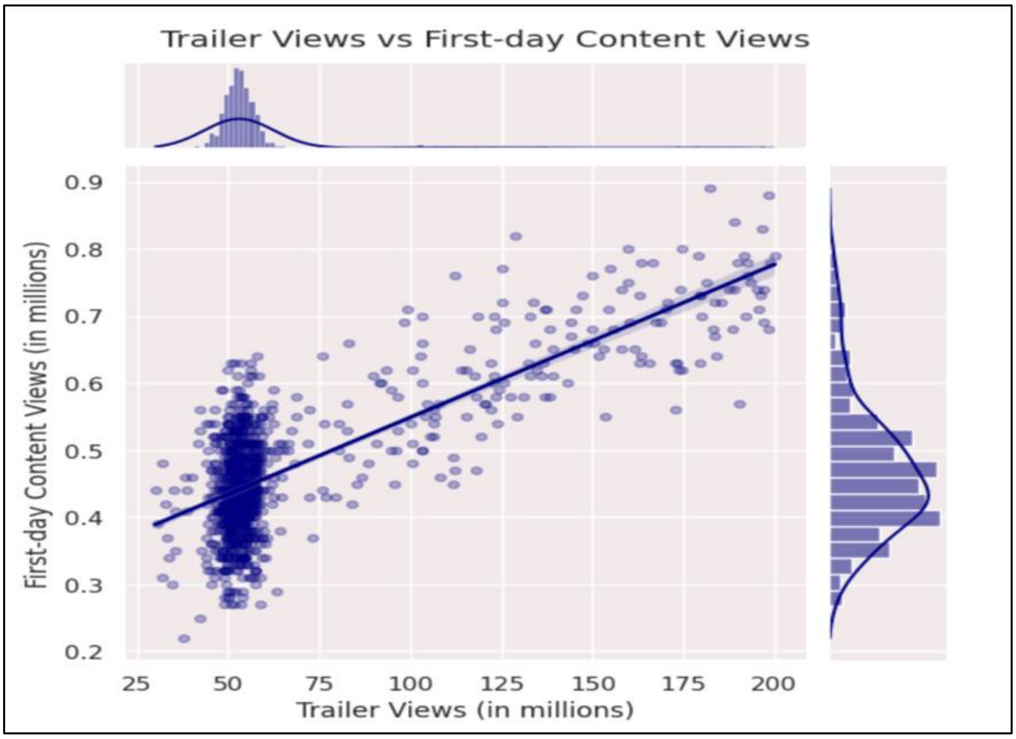

Figure 17. Distribution of views_trailer vs. views_content.

Observations

- The scatter points exhibit a clear upward trend, indicating a strong positive linear relationship because of Correlation (r) = ~0.75M (from the heatmap data above).
- The data clusters around ~50-55 million trailer views and 0.3-0.5 million content views.
- The regression line indicates a positive slope, demonstrating that higher trailer views correlate with higher first-day content views.
- This means that as the number of trailer views increases, so do first-day content views.

- Therefore, promoting trailers effectively can dramatically improve content performance upon debut.

2.3 Multivariate Analysis

Distributions of all columns by genre:

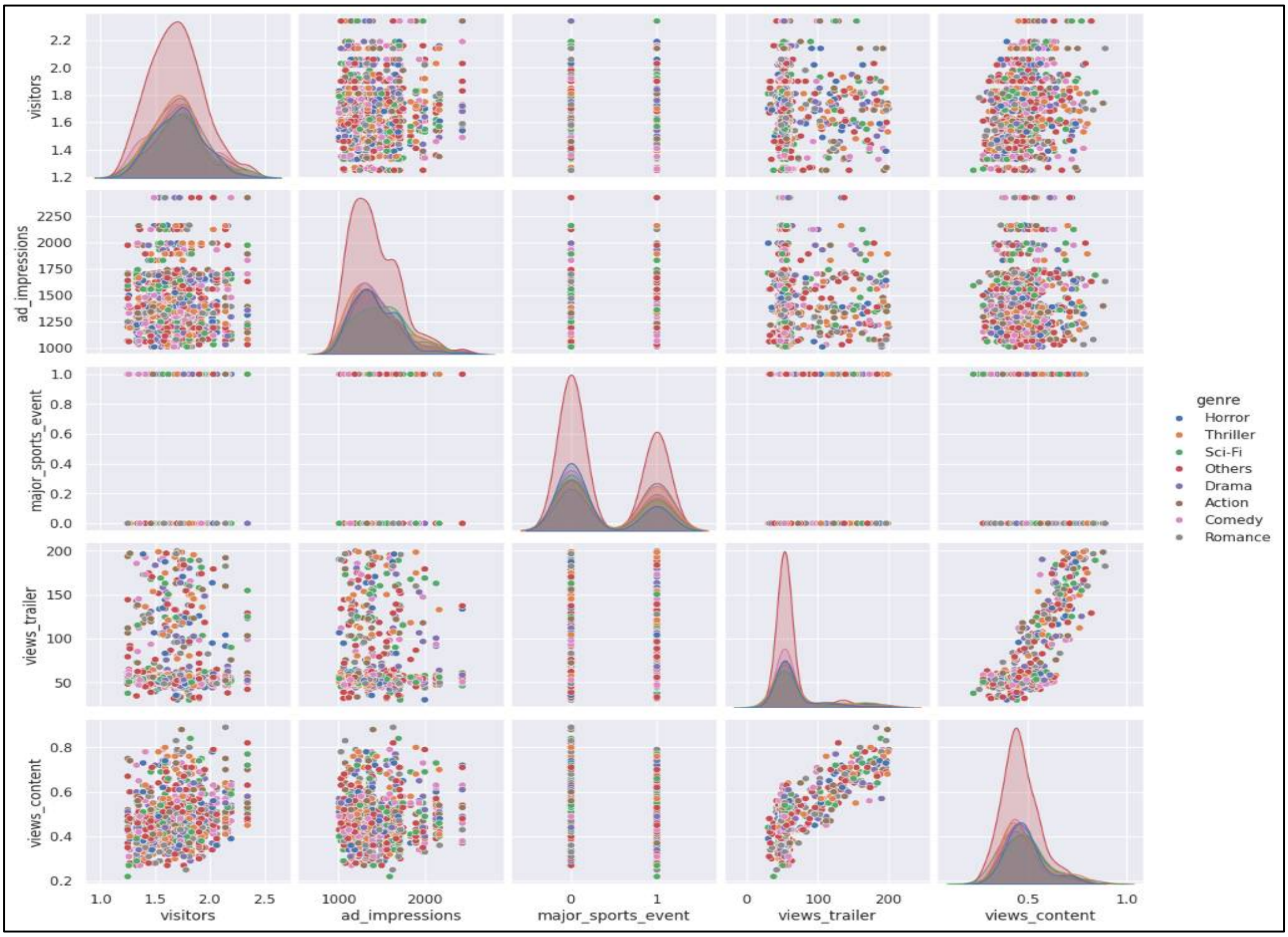

Figure 18. Distributions of all columns by genre.

Observations by genre

1. views_trailer vs. views_content

- Sci-Fi and Action genres are clustered on the top right: high trailer buzz leads to higher content views (70M trailer views and 0.55 million content views).
- Romance and comedy are primarily viewed in the bottom-left corner, with lower trailer views and correspondingly lower content views.
- Drama, Others, and Thriller are spread over the range, with no clear clustering.
- Trailer view is a powerful performance driver, particularly in genres such as Sci-Fi and Action. Genres with little trailer engagement perform poorly on launch day.

2. ad_impressions vs. views_content

- Regardless of performance level, all genre colours have a high degree of overlap, with 1.2 billion to 1.6 billion ad impressions.
- No clear vertical division is noted between genres.
- Ad impressions alone are not a significant predictor of genre performance. * Marketing may spend evenly without regard to genre.

3. visitors vs. views_content

- This pair doesn't focus on any one genre.
- The distribution of data indicates that visitor volume is consistent across genres.
- Platform-wide visitor statistics are steady and non-genre specific.
- This shows that visitor count does not explain genre-specific variation.

4. major_sports_event vs. views_content

- Because major_sports_event is binary, its columns exhibit no significant genre bias.
- Every genre has points on and off sports event days.
- Content is not released around sports events based on genre.
- Scheduling appears to be independent of event presence.

5. Genre Slope Patterns

- The trailer-content view relationship has a positive slope across all genres, as seen without regression lines.
- The intercept shifts, that is, genres like as sci-fi and action receive more content views at the same trailer level.
- These genres have a higher conversion rate of trailer views to content views, maybe due to strong fan communities.

2.4 Overall Insights from EDA

1.Key Performance Indicators

The Effect of views_trailer

- There is a **strong positive correlation (0.75)** between trailer views and first day content views.
- Engagement with trailers is the **most consistent driver** for predicting content performance.
- **Sci-fi and Action** genres have better performance in trailer-to-content conversion rates compared to Romance and Comedy genres which lag behind in both trailer engagement and conversion.

Timing the Release

- ***Day of the Week:***
  - Content is released mainly on **Fridays and Wednesdays which constitute 70% of all content releases**.
  - **Saturday has the highest content performance** metrics with **0.498 million** average views per content.

    - **Friday has the lowest content performance** at **0.447 million** views per content, although it has the highest number of releases.
    - On **Monday and Tuesday there is a clear underutilization** trend at 2.3-2.4% of releases.
- ***Viewership by season:***
    - **Summer continues to be the most popular** season for viewership with an average of **0.497M**.
    - **Winter season is consistent** with winter viewership averaging at **0.485M**.
    - **Fall is noticeably the weakest season** for viewership at **0.445M**.
    - There is an **11% difference in performance between summer and fall**.

2.Genre Performance Metrics

Best Performing Genres

- **Sci-Fi, Horror, and Thriller** have the highest average first-day viewership (0.47-0.49M).
- The **Action genre** captures the most unique platform users (1.74M average).
- These genres show better engagement as well as stronger fan activation and conversion ratios.

Genre Distribution

- The **"others" category dominates** distribution with 26% share of the total content released.
- **Balanced distribution** across standard genres, each falling within the 10.1 - 11.4% range.
- **Concentration of content** in unclassified categories.

3.External Influences

Impact of Sports Events

- **Major sport event drops the viewership** around 10% from 0.49 Million to 0.44 million.
- There is a **moderate negative correlation** (r = -0.24) with performance on first day.
- Overall platform traffic and ad impressions remain unaffected by sporting events.

4.User Involvement & Platform Activities

Content Consumption

- The **average first-day view number is around 0.47M** has a moderate right-skew distribution.
- Some **users are able to achieve 0.7-0.9M views** indicating potential for virality.
- Most content shows **reliable engagement trends** ranging from mid-concentration to mid-high concentration.

Marketing Efficiency

- **ad impressions shows little impact** on success of content with weak covariation (r = 0.05).
- **Trailer views are 15 times more predictive** than ad volume for content performance.

Platform Engagement Trends

- The number of **daily visitors is consistent** with an average of 1.7 million and very low variance of 0.23 Million.
- Success of content remains unattainable with **high variability of ad impressions** (SD=289.5).
- Content performance fails to strongly predict platform traffic as it has **weak correlation with platform traffic** (r=0.26).

5.Distribution Insights

Content Release Strategy Patterns

- **Seasonal timing bias** for mid-week and weekend transitions.
- **Season is almost uniformly distributed** around a all seasons (24.4-25.7%).
- **Weekend releases have more per-content engagement** than weekday volume.

Audience Behaviour

- **Stable platform usage** by visitors regardless of content strategy.
- **Clearly defined* genre preferences** with measurable performance gaps.
- **External event sensitivity** affects content consumption behaviour.

2.5 Answering Key Questions

Q1. **What does the distribution of content views look like?**

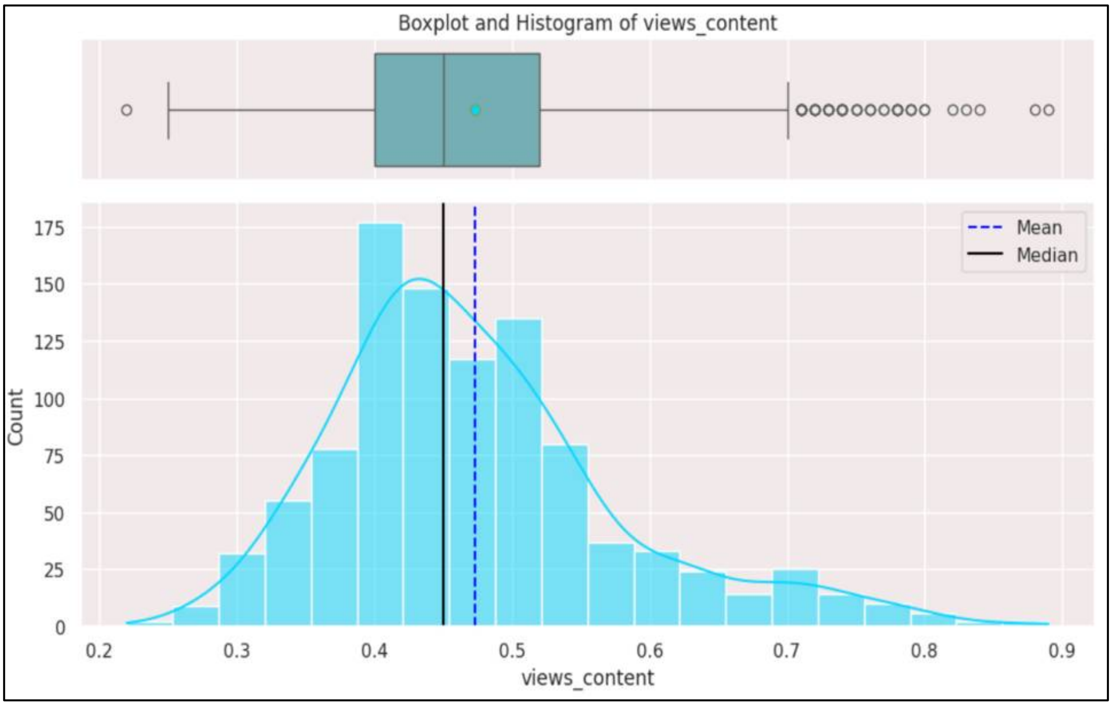

Figure 19. Distribution for Key Q.1

Observations

Histogram:

- **The distribution of first day content view has moderate right-skewness and approximately normal, with majority of content views is concentrated between 0.4M to 0.5M.**
- First-day content views with Mean = 0.47 and Median = 0.45M.

Boxplot:

- The boxplot reveals several outliers on the beyond upper whiskers with 0.7-0.9 million view on first-day of content release and one outlier outside the lower whiskers with approximately 0.2 million views on first-day of content release, potentially due to popularity, strong marketing, or other factors.
- Overall, the content views data is relatively normally distributed with a slight right skew, indicating a typical range of content views values around the mean.
- Analysing the relationship between trailer views and first-day content views can reveal whether good trailer promotion influences overall viewership.

Q2. What does the distribution of genres look like?

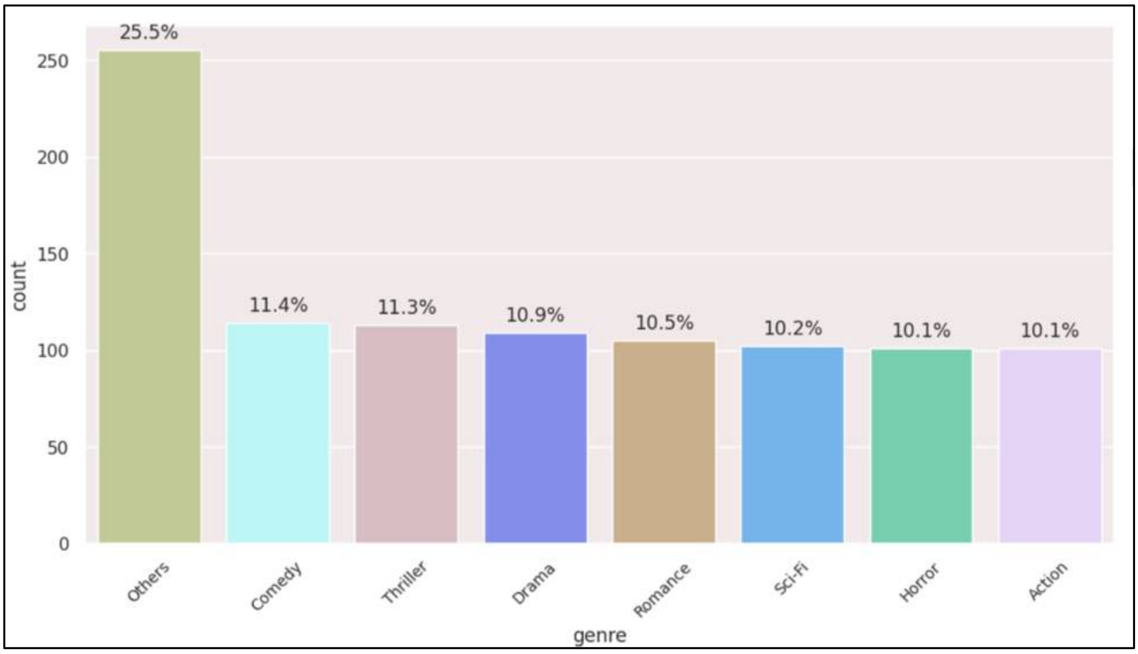

Figure 20. Distribution for Key Q.2

Observations

- The bar chart represents a long-tail catalogue with unbalanced distribution.
- "Others" genre takes the lead with 26% of content release in this category, followed by Comedy (11.4%), Thriller (11.3%),Drama (10.9%) and Romance (10.5%).
- Others genre might represent significant content that doesn't fit traditional genres.
- The amount of content releases in the action, horror, and science fiction genres is relatively low (10.1-10.2%).
- Understanding the popularity of various genres might be useful in content strategy.

Q3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

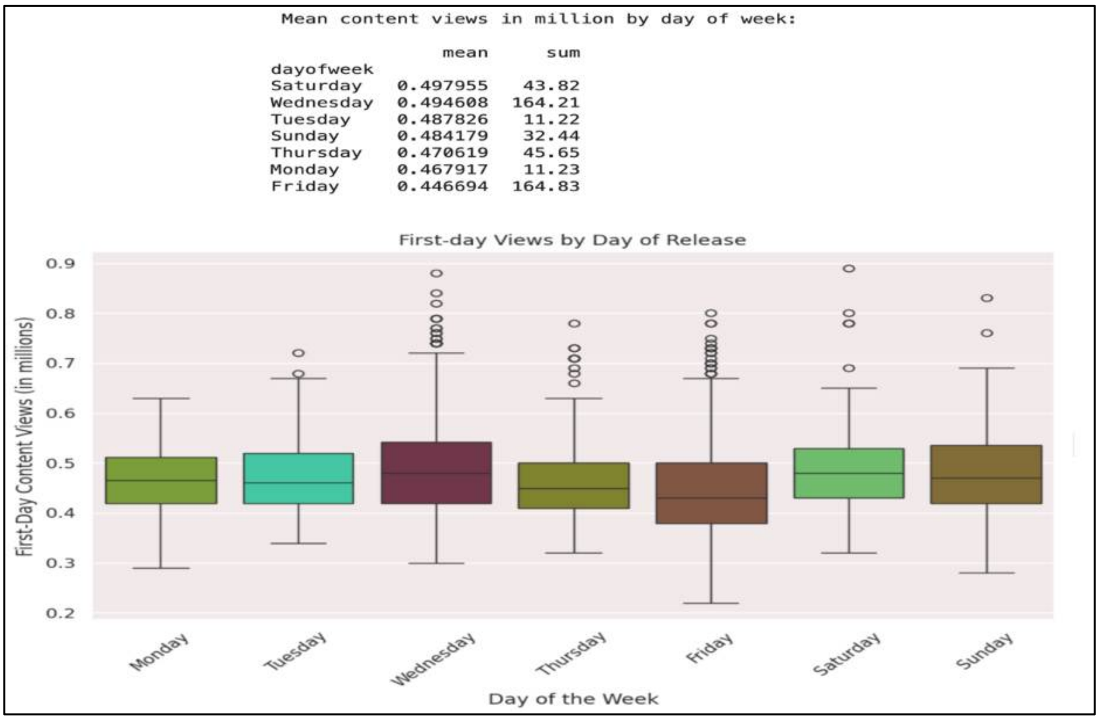

Figure 21. Distribution for Key Q.3

Observations

- In the boxplot, Saturday has the highest median and the highest average views per content (mean = 0.498M).
- Wednesday exhibits the most variety with several high-performing outliers and ranks closely in mean views (0.495M).
- Despite having the most views overall, Friday has the lowest average views per piece of content (0.447M), suggesting a large volume of releases.
- The fact that Monday and Tuesday have the fewest total views suggests that there may be fewer content drops on those days.
- Wednesday and Friday are the days with the most outliers, suggesting occasional viral content, leading to high first-day viewership.

Interpretation

- Viewership fluctuates dramatically each day of the week.
- While Friday has the most releases (total views), it underperforms per-content, indicating content saturation or audience fatigue.
- Saturday is the best day to optimise per-content performance, both in terms of average and boxplot distribution (high median and moderate spread).
- Wednesday is a good midweek candidate because of its high average and frequent spikes, which are probably caused by targeted content.
- Releasing significant or high-ticket content on Saturday or Wednesday may increase engagement.
- Friday may be set aside for larger, lower-priority releases in which performance is less crucial.

Conclusion

- The day of release has a distinct and measurable effect on first-day content viewership. A data-driven scheduling strategy can greatly boost first-day content views performance on the platform.

Q4. How does the viewership vary with the season of release?

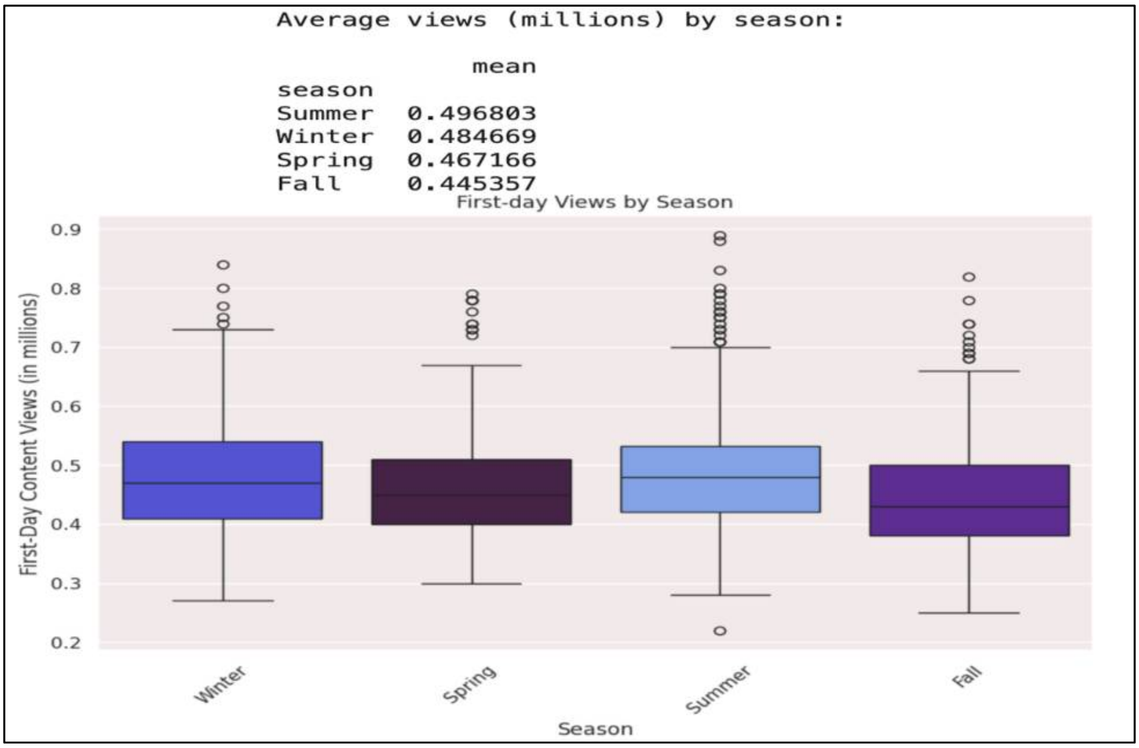

Figure 22. Distribution for Key Q.4

Observations

Summer

- Highest median viewership (0.45M) and mean viewership (0.497M).
- Outliers in both ends of whiskers indicate breakout potential (up to 0.7M+).

Winter

- Most consistent performance (narrow IQR). 2.3% below Summer median but reliable with 2nd highest mean viewership (0.485M).
- There are outliers on the upper whiskers in the range of 0.73M-0.85M.

Spring

- Spring ranks third in terms of mean (0.467M), with an IQR similar to Winter but a lower median.
- Outliers are present in upper whiskers of viewership range ~(0.71M-0.8M).

Fall

- Fall has the lowest mean (0.445M) and median, with a slightly narrower range that centres lower than other seasons.
- Outliers on upper end with viewership in range ~(0.65M-0.82M).

Summer and **Winter** have high outliers (≥ 0.85 million), indicating infrequent blockbuster releases.

Interpretation Seasonality affects average first-day views, which vary by around 11% between peak (Summer) and low (Fall) seasons.

Summer advantage: Increased leisure time (school breaks, vacations) is likely to improve engagement, making summer the best period for premium releases.

Winter opportunity: Winter's mean is similar to summer's, implying that vacation breaks and indoor entertainment fuel increased viewership.

Lower averages may indicate conflict with holiday preparations or new TV seasons. Reserve Autumn for lower-priority programs or allocate extra marketing.

Conclusion

- Thus, there's also a noticeable difference in the distribution of viewership across different seasons.
- Viewership is highest in the summer and lowest in the fall, with winter and spring in between.
- This suggests that viewers might be more inclined to watch content on ShowTime during Summer, Winter seasons, potentially due to factors like weather or seasonal preferences.
- To maximise opening-day impact, schedule flagship titles in Summer or Winter, and increase promotion for Fall releases to compensate for seasonal headwinds

Q.5 What is the correlation between trailer views and content views?

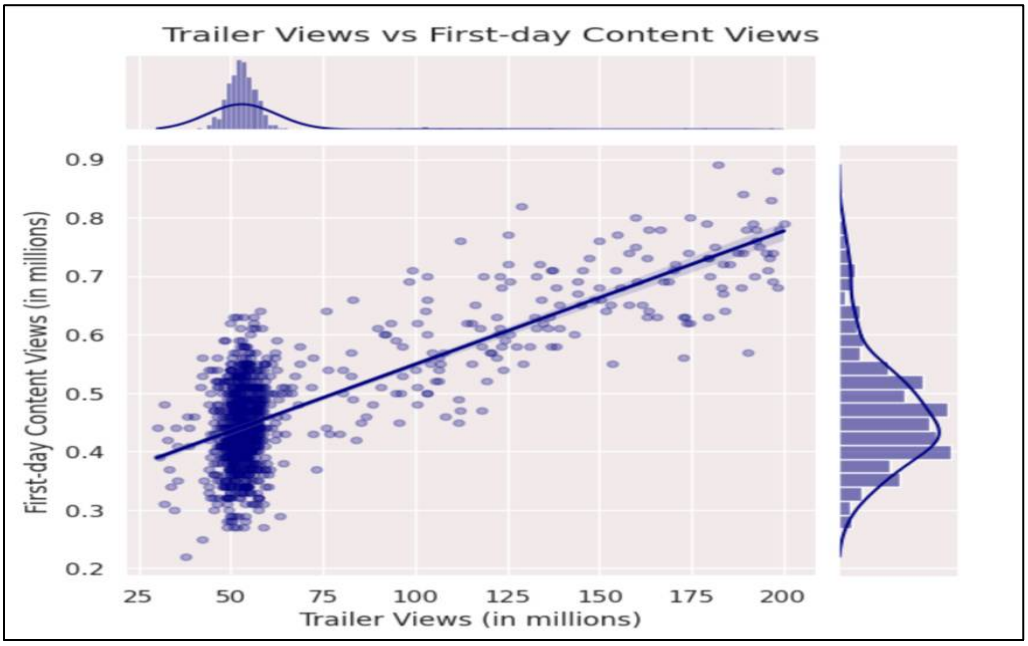

Figure 23. Distribution for Key Q.5

Observations

- The scatter points exhibit a clear upward trend, indicating **a strong positive linear relationship because of Correlation (r) = ~0.75M (from the heatmap data above).**
- The data clusters around ~50-55 million trailer views and 0.3-0.5 million content views.
- The regression line indicates a positive slope, demonstrating that higher trailer views correlate with higher first-day content views.
- **This means that as the number of trailer views increases, so do first-day content views.**
- **Therefore, promoting trailers effectively can dramatically improve content performance upon debut.**

3 DATA PROCESSING

3.1 Data Cleaning

Missing Values:

Missing Values:

	0
visitors	0
ad_impressions	0
major_sports_event	0
genre	0
dayofweek	0
season	0
views_trailer	0
views_content	0

dtype: int64

Table 5. Number Of Missing Values.

- There are **no missing values** in the dataset.
- **This ensured that no Data Imputation or Removal techniques were required, thereby preserving the data integrity of the original dataset.**

Duplicate Values:

- As observed, there are **no duplicate values** in the dataset**.**
- **This suggests that all data are unique, ensuring data integrity and reliability for further analysis.**

3.2 Outliers Detection & Treatment

Outlier Detection Using Boxplot:

Outlier detection will not done on the major_sports_event variable because it is a binary indication (0 = no, 1 = yes). Outlier detection technique, such as IQR, is only relevant to continuous numerical variables with a large range of values. Because major_sports_event only accepts two valid values, it has no statistical outliers by definition.

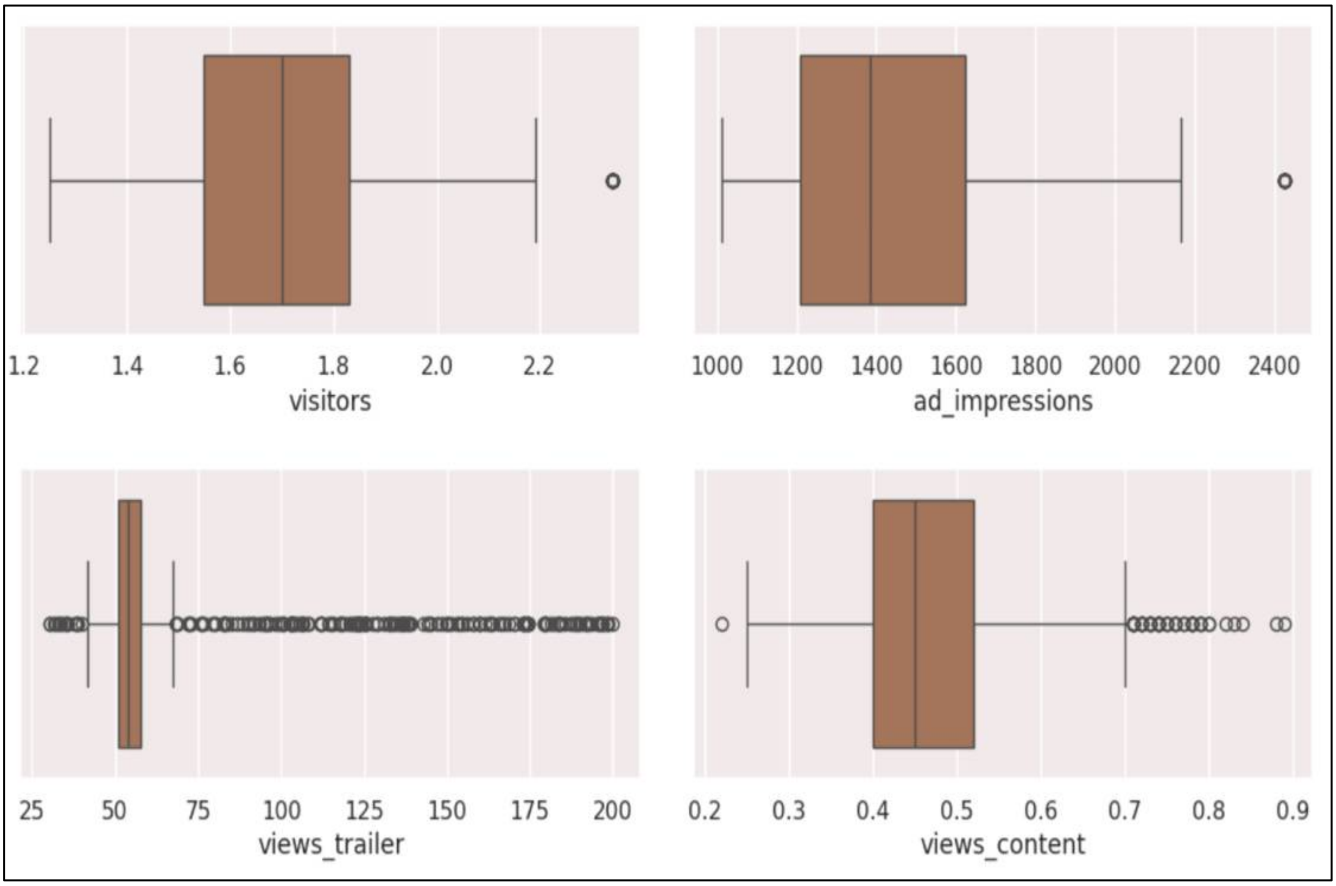

Figure 24. Box plot for Outlier Detection.

Observations

- ***visitors:*** It has one outlier above upper end with range above 2.2M.
- ***ad_impressions:*** It has one outlier on upper whiskers having ad impression more than 2400M.
- ***views_trailer:*** It had several outliers on outside of both upper (75M-200M) and lower whiskers (less than 50M).
- ***views_content****:* It also has many outliers but less compared to trailer views. Lower end has one outlier with views less the 0.35M, while upper whiskers outlier range from 0.7M-0.9M approximately.

Outlier Detection Using IQR Method & Treatment Technique :

	lower_bound	upper_bound	outlier_count	outlier_percentage
visitors	1.13	2.25	20.0	2.0
ad_impressions	590.32	2243.68	13.0	1.3
views_trailer	40.74	67.97	189.0	18.9
views_content	0.22	0.70	47.0	4.7

Table 6. Outlier Detection Using IQR Method.

Observations

visitors

- It has **2% outliers** i.e. (20/1000).
- Extreme visitor counts could distort relationship with viewership. Since, visitors outlier percentage (2%) is low.
- Therefore, we will use capping (**Winsorizing**) technique to treat these outliers as it preserve the data without deleting it.

ad_impressions

- It has **1.3% outliers** i.e. (13/1000).
- ad_impressions outliers may represent data errors or special campaigns.
- Extreme values may disproportionately influence regression model coefficients.
- Therefore, **we will use Winsorize (capping) technique to treat the outliers** because minimal data loss and it maintains realistic ad impressions range while reducing skewness

views_trailer

- It has **18.9% outliers** i.e. (189/1000) which is high outlier percentage.
- This high outlier count makes capping/removal problematic.
- Therefore, **we will use Log transformation technique to treat the outliers** as it reduces skew while retaining all data points.
- This technique also ensures linear regression assumptions (normality) are met

views_content(target variable)

- Views_content has **4.7% outliers** i.e. (47/1000).
- These **outliers could represent genuine high/low viewership events** (e.g., viral hits or flops).
- Removing them would bias the model by ignoring critical real-world scenarios.
- Any transformations on the target can distort predictions.
- Therefore, **we will not treat this outliers for views_content variable**.

Verification Post - Outlier Treatment:

- **visitors:** 0.0% outliers remaining after capping.
- **ad_impressions:** 0.0% outliers remaining after capping.
- **views_trailer:** 17.0% outliers remaining after log transformation.

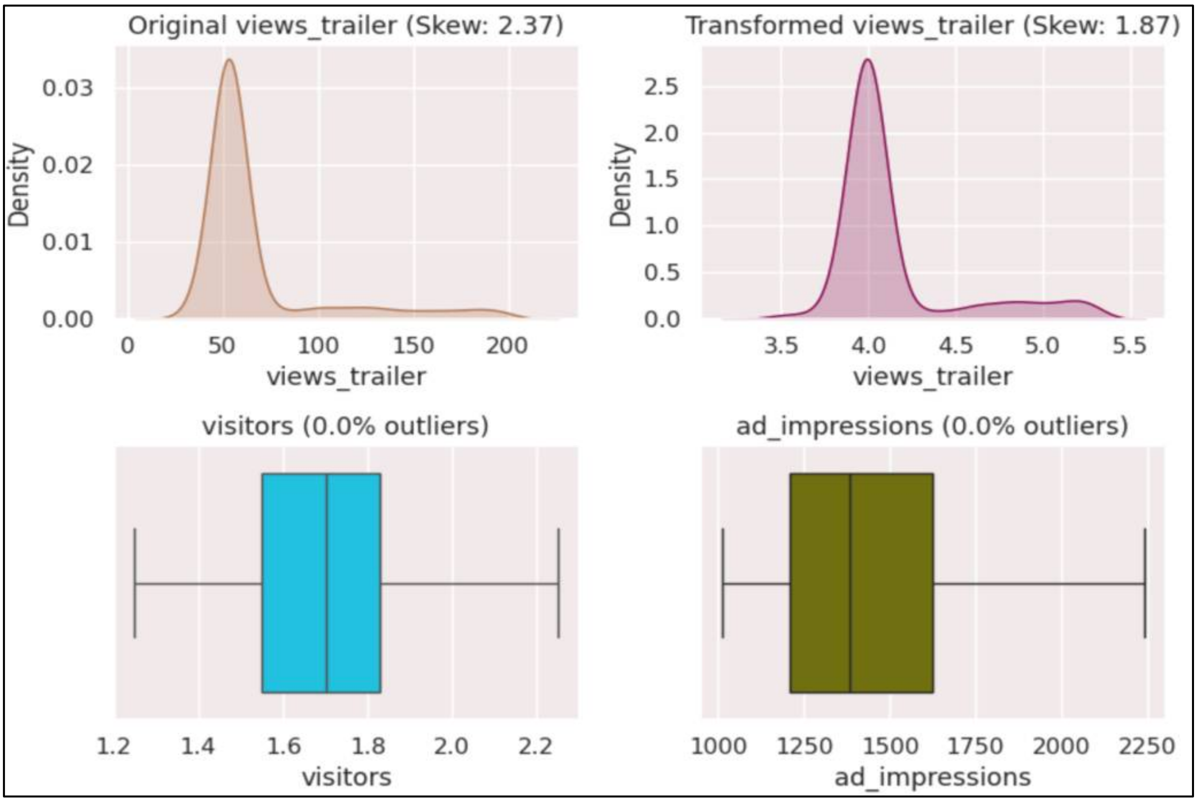

Figure 25. Verification Post - Outlier Treatment.

Observations

- We can clearly see that there are no remaining outliers for visitors and ad_impressions from statistical and visual evidences.
- We can also see there is reduced outliers for views_trailer from 18.9% to 17%. These remaining outliers are business outliers and are in optimal range now.

Insights

Success Criteria Met:

- Winsorizing completely resolved outliers for two features (visitors and ad_impressions).
- Log transformation reduced skewness in views_trailer by approximately 50% (a normal outcome).

views_trailer 17% remaining outlier is okay because:

- These are no longer statistical outliers, but business outliers: trailers that truly performed exceptionally well or poorly.
- Their reduced magnitude upon transformation minimises impact on the model.
- Removing them would lose key success and failure situations.

3.3 Feature Engineering

Creating new feature in DataFrame:

We know that in genre **'Others' has unspecified categories** (26% of data) and it could dominate or obscure patterns, and may create multicollinearity issue. We will consider this for feature engineering to avoid such issues. Instead of removing this data, **we create a new binary variable other_genre** that:

- **Takes the value 1 if the genre is "Others"**
- **Takes the value 0 otherwise**

This allows the model to assess whether being part of this ambiguous category impacts viewership, without distorting the rest of the genre features.

Top 5 rows of updated DataFrame with new feature

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content	other_genre
0	1.67	1113.81	0	Horror	Wednesday	Spring	4.055257	0.51	0
1	1.46	1498.41	1	Thriller	Friday	Fall	3.983227	0.32	0
2	1.47	1079.19	1	Thriller	Wednesday	Fall	3.906809	0.39	0
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	3.928093	0.44	0
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	4.040064	0.46	0

Table 7. Updated Df With New Feature.

- We can see new feature i.e. "other_genre" is created in our dataset.

Value Counts Of New Feature:

other_genre	count
0	745
1	255

dtype: int64

Table 8. New Feature Value Counts.

Observations

- 255 counts (25.5%) belong to the "Others" genre
- 745 counts (74.5%) are specific genres (Actions, Horror, Thriller, etc.)

3.4 Data Preparation For Modeling

One-Hot Encoding:

The categorical columns (‘dayofweek’, ‘genre’ and ‘season’) converted to numerical columns using One-Hot Encoding.

other_genre	genre_Comedy	genre_Drama	genre_Horror	genre_Romance	...	genre_Thriller	dayofweek_Monday	dayofweek_Saturday	dayofweek_Sunday	dayofweek_Thu
0	0	0	1	0	...	0	0	0	0	
0	0	0	0	0	...	1	0	0	0	
0	0	0	0	0	...	1	0	0	0	
0	0	0	0	0	...	0	0	0	0	
0	0	0	0	0	...	0	0	0	1	

Table 9. Dummy Variables.

Checking Missing Values In Dummy Variables:

other_genre	0
genre_Comedy	0
genre_Drama	0
genre_Horror	0
genre_Romance	0
genre_Sci-Fi	0
genre_Thriller	0
dayofweek_Monday	0
dayofweek_Saturday	0
dayofweek_Sunday	0
dayofweek_Thursday	0
dayofweek_Tuesday	0
dayofweek_Wednesday	0
season_Spring	0
season_Summer	0
season_Winter	0

Table 10. Dummy Variable Missing Value Count.

- There are **no missing values** in the dummy variables columns.
- **This ensured that no Data Imputation or Removal techniques were required for dummy variables columns, thereby preserving the data integrity of the original dataset.**

4 MODEL BUILDING – LINEAR REGRESSION

4.1 Original Model

Before model building:

- In this section initially we selected our **target variable** (**views_content**) from the DataFrame and separated it from the independent variables(predictors).
- Then we added an intercept value (constant value).
- **We split the data into train and test set:**
  - x_train → 70% of feature data used to train the model
  - x_test → 30% of feature data used to test (evaluate) the model
  - y_train → Target values for x_train
  - y_test → Target values for x_test

Shape of Train & Test dataset:

- Number of rows in train data = **700**
- Number of rows in test data **= 300**

Building Original Model using Linear Regression:

Here, we build the Model using sm.ols() with x_train and y_train as input and fitted the model using fit().

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.783
Method:                 Least Squares   F-statistic:                     126.9
Date:                Wed, 09 Jul 2025   Prob (F-statistic):         1.01e-213
Time:                        10:03:25   Log-Likelihood:                 1120.1
No. Observations:                 700   AIC:                            -2198.
Df Residuals:                     679   BIC:                            -2103.
Df Model:                          20
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     -0.6983      0.029    -24.259      0.000      -0.755      -0.642
visitors                   0.1298      0.008     15.991      0.000       0.114       0.146
ad_impressions           4.24e-06   6.82e-06      0.622      0.534   -9.15e-06    1.76e-05
major_sports_event        -0.0630      0.004    -15.841      0.000      -0.071      -0.055
views_trailer              0.2211      0.005     41.893      0.000       0.211       0.231
other_genre                0.0058      0.007      0.813      0.416      -0.008       0.020
genre_Comedy               0.0087      0.008      1.087      0.277      -0.007       0.024
genre_Drama                0.0101      0.008      1.232      0.218      -0.006       0.026
genre_Horror               0.0083      0.008      1.015      0.310      -0.008       0.024
genre_Romance              0.0012      0.008      0.139      0.890      -0.015       0.018
genre_Sci-Fi               0.0162      0.008      1.962      0.050   -1.13e-05       0.032
genre_Thriller             0.0080      0.008      0.989      0.323      -0.008       0.024
dayofweek_Monday           0.0309      0.012      2.601      0.010       0.008       0.054
dayofweek_Saturday         0.0570      0.007      7.912      0.000       0.043       0.071
dayofweek_Sunday           0.0366      0.008      4.639      0.000       0.021       0.052
dayofweek_Thursday         0.0157      0.007      2.301      0.022       0.002       0.029
dayofweek_Tuesday          0.0250      0.014      1.812      0.070      -0.002       0.052
dayofweek_Wednesday        0.0460      0.005     10.168      0.000       0.037       0.055
season_Spring              0.0249      0.005      4.618      0.000       0.014       0.035
season_Summer              0.0456      0.005      8.311      0.000       0.035       0.056
season_Winter              0.0271      0.005      5.048      0.000       0.017       0.038
==============================================================================
Omnibus:                        5.231   Durbin-Watson:                   1.975
Prob(Omnibus):                  0.073   Jarque-Bera (JB):                5.162
Skew:                           0.172   Prob(JB):                       0.0757
```

Table 11. Original Model.

Original Model Coefficients:

	Feature	Coefficient
0	const	-0.698283
1	visitors	0.129802
2	ad_impressions	0.000004
3	major_sports_event	-0.062978
4	views_trailer	0.221061
5	other_genre	0.005775
6	genre_Comedy	0.008727
7	genre_Drama	0.010071
8	genre_Horror	0.008342
9	genre_Romance	0.001179
10	genre_Sci-Fi	0.016236
11	genre_Thriller	0.008028
12	dayofweek_Monday	0.030935
13	dayofweek_Saturday	0.056961
14	dayofweek_Sunday	0.036563
15	dayofweek_Thursday	0.015655
16	dayofweek_Tuesday	0.025005
17	dayofweek_Wednesday	0.045973
18	season_Spring	0.024863
19	season_Summer	0.045565
20	season_Winter	0.027072

Table 12. Original Model Coefficients.

Interpretation of Original Model Regression Results:

R-squared:-

- In our case value is 0.789.
- It means that 78.9% of the total variance in views_content is explained by the specific independent variables in the model.

Adjusted. R-squared:- It reflects the fit of the model.

- Adjusted R-squared values generally range from 0 to 1, where a higher value generally indicates a better fit, assuming certain conditions are met.
- In our case, the value for adj. R-squared is **0.783**, which is good, suggest good fit model.
- A result of 0.783 shows that the model explains 78.3% of the variation in first-day viewing, even after controlling for various factors. This shows an excellent model fit.

Const. coefficient:- It is the Y-intercept.

- It means that if all the predictor variable coefficients are zero, then the expected output (i.e., Y) would be equal to the *const.* coefficient.
- In our case, the value for *const*. coefficient is **-0.6983.**
- Since, this value is negative, it has no direct practical meaning in this context (all predictors being zero is implausible), yet it is mathematically required for the model.

F-statistic:-

- F-statistic = **126.9**
- Probability (F-statistic): 1.01e-213.
- This demonstrates that the entire model is statistically significant.
- The very low p-value implies that at least one predictor variable is significantly associated with the response variable.

Coefficient of a predictor variable:- It represents the change in the output Y due to a change in the predictor variable (everything else held constant). We have total of 20 features in this model. Example:

- In our case, the coefficient of **views_trailer** is **0.2211**. This means for every additional 1 million trailer views raises the first-day view total by an average of 0.221 million (221k).
- For the coefficient of '***visitors***' is **0.1298**. This means for every 1 million additional platform visitors, there are around 130k more first-day content views.
- For the coefficient of **major_sports_event** is **-0.0630**, suggesting launching content during a sporting event reduces views by around 63k.
- For **genre_Sci-Fi** with coefficient value of **0.0162**, indicates Sci-Fi content performs somewhat better than the base genre (Action), resulting in around 16k extra views.
- For the coefficient of **dayofweek_Saturday** is **0.0570**, shows Saturday releases generate ~57k more views than Friday (baseline).
- Other categorical factors (days and seasons) have similar interpretations, with significant ones (p-value < 0.05) having a considerable impact on viewing.

Non-significant Variables:-

- **ad_impressions, other_genre, and most genre dummies have strong p-values (p > 0.05)**, indicating that they have **no statistically significant effect on first-day views** when other factors are held constant.
- This could be due to overlap or multicollinearity with more significant variables, such as views_trailer.

Original Model Performance Check:

We checked the performance of the original model using different metrics.

- We used metric functions defined in sklearn for RMSE, MAE, and R^2.
- We defined a function to calculate MAPE and adjusted R^2.
- The mean absolute percentage error (MAPE) measures the accuracy of predictions as a percentage, and can be calculated as the average absolute percent error for each predicted value minus actual values divided by actual values. It works best if there are no extreme values in the data and none of the actual values are 0.
- We created a function which will printed out all the above metrics in one go.
- We evaluated the model's performance on both train and test dataset.

	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
Train	0.048844	0.038426	0.788913	0.782374	8.549048
Test	0.050487	0.040592	0.767519	0.749958	9.009553

Original Model Performance:

Figure 26. Original Model Performance Metrics.

Observations:

RMSE (Root Mean Squared Error)

- **Train = 0.0488**
- **Test = 0.0505.**
- The target variable (views_content) spans from 0.2M to 0.9M, resulting in an average prediction error of roughly 50,000 views (RMSE ~ 0.05).
- The acceptable range for RMSE in this situation is <0.05 = Excellent, 0.05-0.10 = Good.
- Our numbers are in the **Excellent-Good range**.

Mean Absolute Percentage Error (MAPE)

- **Train MAPE = 8.549(8.55%)**
- **Test MAPE = 9.009 (9.01%)**.
- This indicates that, on average, our predictions and the actual results differ by about 9%.
- Our model fulfils the best-practice standard, with an **acceptable MAPE range of <10% (excellent)**.

R-squared and adjusted R-squared

- **R-squared Train** = **0.789**
- **R-squared Test** = **0.767**
- This indicates our approach accounts for roughly **76-79% of the variation in first-day content views.**
- Adjusted R^2 reflects the number of predictors:
- **Adjusted R^2 Train = 0.782**
- **Adjusted R^2 Test = 0.749**
- The slight drop from train to test is normal and acceptable.
- Acceptable R^2 varies by context, however with business data, **$R^2 > 0.75$ is regarded strong**.

Conclusion:

- So far, Model fits well, as evidenced by its performance on the test set, which is highly similar to the training set. But we still need to test the assumptions.
- **No considerable overfitting** is detected.
- Errors are low and within acceptable limits.
- This suggests that the model is resilient, dependable, and capable of making predictions based on previously unseen data for far.

5 TESTING ASSUMPTIONS FOR LINEAR REGRESSION MODEL

We will be checking the following Linear Regression assumptions:

1. **No Multicollinearity**
2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

5.1 Test for Multicollinearity

Multicollinearity occurs when predictor variables in a regression model are correlated. We used Variance Inflation Factor (VIF) for detecting (or testing) multicollinearity on train dataset for each feature. We used variance_inflation_factor() to calculate the VIF values.

- If VIF is between 1 and 5, then there is low multicollinearity.
- If VIF is between 5 and 10, we say there is moderate multicollinearity.
- If VIF is exceeding 10, it shows signs of high multicollinearity.

feature	VIF
const	235.813072
other_genre	2.573661
genre_Drama	1.925017
genre_Thriller	1.920533
genre_Comedy	1.917612
genre_Horror	1.903682
genre_Sci-Fi	1.864272
genre_Romance	1.753434
season_Winter	1.569560
season_Summer	1.566865
season_Spring	1.541060
dayofweek_Wednesday	1.316002
dayofweek_Thursday	1.170615
dayofweek_Saturday	1.155980
dayofweek_Sunday	1.150704
major_sports_event	1.067316
dayofweek_Monday	1.063398
dayofweek_Tuesday	1.062225
ad_impressions	1.029726
visitors	1.026610
views_trailer	1.024779

Table 13. Features VIF Values.

Observations

- **There is no significant multicollinearity in the model**, as seen by the VIF values of all predictor variables (except from constant), are much below the VIF <= 5
- Therefore, the **no multicollinearity assumption is satisfied**.

Dealing With High P Values:

Some of the dummy variables in the data had p-value > 0.05. So, they were not significant and we dropped them one at a time as sometimes p-values change after dropping a variable. Instead, we did the following:

- Build a model, check the p-values of the variables, and drop that variable from your dataset.
- Re-fit the model using the updated dataset (with the dropped column).
- Repeat the above two steps till there are no columns with p-value > 0.05

```
Dropping 'genre_Romance' with p-value 0.8897
Dropping 'ad_impressions' with p-value 0.5354
Dropping 'other_genre' with p-value 0.3765
Dropping 'genre_Thriller' with p-value 0.4600
Dropping 'genre_Horror' with p-value 0.5319
Dropping 'genre_Comedy' with p-value 0.5275
Dropping 'genre_Drama' with p-value 0.4434
Dropping 'genre_Sci-Fi' with p-value 0.0888
Dropping 'dayofweek_Tuesday' with p-value 0.0618
All remaining features are significant (p ≤ 0.05)
```

Figure 27. Dropped Features with P Value > 0.05

Now we will be preparing refined train and test datasets to build a updated regression model with only significant features that are left after we dropped the bad features. We will perform assumptions test on this updated model.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.786
Model:                            OLS   Adj. R-squared:                  0.783
Method:                 Least Squares   F-statistic:                     229.7
Date:                Wed, 09 Jul 2025   Prob (F-statistic):          8.48e-222
Time:                        10:03:25   Log-Likelihood:                 1115.3
No. Observations:                 700   AIC:                            -2207.
Df Residuals:                     688   BIC:                            -2152.
Df Model:                          11
Covariance Type:            nonrobust
======================================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------
const                 -0.6842      0.026    -26.269      0.000      -0.735      -0.633
visitors               0.1295      0.008     16.029      0.000       0.114       0.145
major_sports_event    -0.0633      0.004    -16.202      0.000      -0.071      -0.056
views_trailer          0.2213      0.005     42.081      0.000       0.211       0.232
dayofweek_Monday       0.0286      0.012      2.420      0.016       0.005       0.052
dayofweek_Saturday     0.0563      0.007      7.885      0.000       0.042       0.070
dayofweek_Sunday       0.0348      0.008      4.472      0.000       0.019       0.050
dayofweek_Thursday     0.0135      0.007      2.006      0.045       0.000       0.027
dayofweek_Wednesday    0.0449      0.004     10.095      0.000       0.036       0.054
season_Spring          0.0248      0.005      4.636      0.000       0.014       0.035
season_Summer          0.0447      0.005      8.310      0.000       0.034       0.055
season_Winter          0.0283      0.005      5.339      0.000       0.018       0.039
==============================================================================
Omnibus:                        4.279   Durbin-Watson:                   1.964
Prob(Omnibus):                  0.118   Jarque-Bera (JB):                4.119
Skew:                           0.162   Prob(JB):                        0.128
Kurtosis:                       3.190   Cond. No.                         66.3
==============================================================================
```

Table 14. Updated Model.

We can clearly after dropping many previous features, updated model build on remaining features.

Performance Check on Updated Model:

Updated Model Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
Train	0.049184	0.038688	0.785965	0.782227	8.603114
Test	0.050850	0.040957	0.764165	0.754304	9.114764

Figure 28. Updated Model Performance Metrics.

Updated Model Observations & Interpretations w.r.t Train Data:

1. R-squared and Adjusted R-squared:

- The **original mode**l had an R-squared of 0.789, while the adjusted R-squared was 0.783, indicating that the model explained approximately 78.3% of the variation in first-day content views.
- In the **updated model** R-squared dropped marginally to 0.786 (after deleting high p-value predictors) and Adjusted R-squared to 0.782.
- The model is now cleaner, with just statistically significant variables remaining, thus this slight decrease (~0.001) is acceptable.
- The model explains approximately 78.6% of the variability in first-day content views.
- Both values are extremely close, showing that there is no overfitting due to insignificant predictors.

2. Model coefficients (significance):

- **All predictors now have p-values < 0.05**, indicating that all variables in the model are statistically significant.
- **Key variables with the biggest influence (high t-values) is views_trailer** (t ≈ 42) is the primary driver of viewership.
- **Predictors: visitors, major_sports_event, dayofweek_Wednesday, and season_Summer all contribute significantly**.

3. Error Metrics:

- ***Original model***
  - The RMSE (Test) is around 0.0504, indicating that predictions are off by approximately 50,400 views on average.
  - MAPE (Test) = 9.01%, means predictions are ~9% off the actuals values.
- ***Updated model***
  - **RMSE (Test) is around 0.0508**. The error is slightly larger, but still acceptable at approximately 50,800 views.
  - **MAPE (Test) = 9.11%**, close to actuals, with an average error of around 9%.
- These little deviations demonstrate that **model accuracy is still maintained even after decreasing predictors.**

5.2 Test For Linearity Of Variables And Independence Of Residuals

- Linearity refers to a straight-line relationship between two variables. Predictor variables must have a linear relationship with the dependent variable.
- The independence of the error terms (or residuals) is crucial. If the residuals are not independent, the confidence intervals for coefficient estimates will be reduced, leading us to wrongly conclude that a parameter is statistically significant.
- Create a plot of fitted values vs residuals. If the residuals do not follow a pattern, the model is linear and independent.
- Otherwise, the model shows evidence of nonlinearity, and the residuals are not independent.
- If the linearity and independence assumptions are not satisfied then we can try to transform the variables and make the relationships linear.

Here, we prepared a DataFrame with Actual values of trained dataset, fitted (predicted) values of train data and residuals (Error) value (Actual- predicted , calculated using resid method), and for plot we used seaborn residplot().

Top 5 Rows of Residuals Values:

	Actual_Values	Predicted_Values	Residuals
731	0.40	0.441486	-0.041486
716	0.70	0.691754	0.008246
640	0.42	0.438357	-0.018357
804	0.55	0.589136	-0.039136
737	0.59	0.547973	0.042027

Table 15. Residuals Values.

Fitted vs. Residuals Plot:

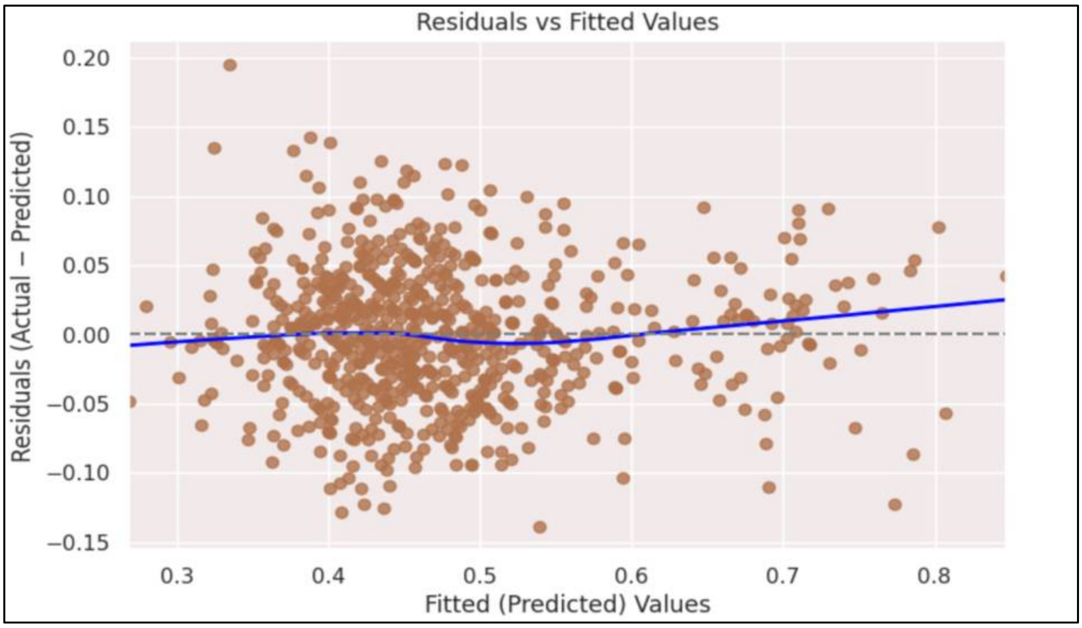

Figure 29. Fitted vs. Residuals Plot.

Observations

- From the plot we can observe that blue points are distributed above and below the dashed grey 0-line, without a distinct systematic curve, and **no specific pattern observed**.
- The red smoothed line remains close to zero throughout the fitted value range. There is a minor upward tilt beyond ≈0.65M, but the difference is minimal (within ~0.02).
- Therefore, **no evidence of non-linearity found**.
- Residuals do not cluster in time-like bands or display any patterns; instead, points appear randomly scattered.
- The independence of residuals appears evident.
- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

5.3 Test For Normality of Residuals

The error terms, or residuals, should follow a normal distribution. If the error components are not normally distributed, the confidence interval for coefficient estimates may become overly large or narrow. When a confidence interval becomes unstable, it makes it harder to estimate coefficients using least squares minimisation. Non-normality indicates that a few unexpected data points must be thoroughly investigated in order to create a better model. Normality check steps:

- The shape of the residuals histogram can provide a first understanding of normality.
- It can also be verified using a Q-Q plot of residuals. If the residuals have a normal distribution, they will result in a straight line plot, otherwise not.
- The Shapiro-Wilk test is another method for determining normality.
- If assumption failed then we can apply transformations such as log, exponential, and arcsinh, etc., to our data.

Histogram Plot of Residuals for Normality check:

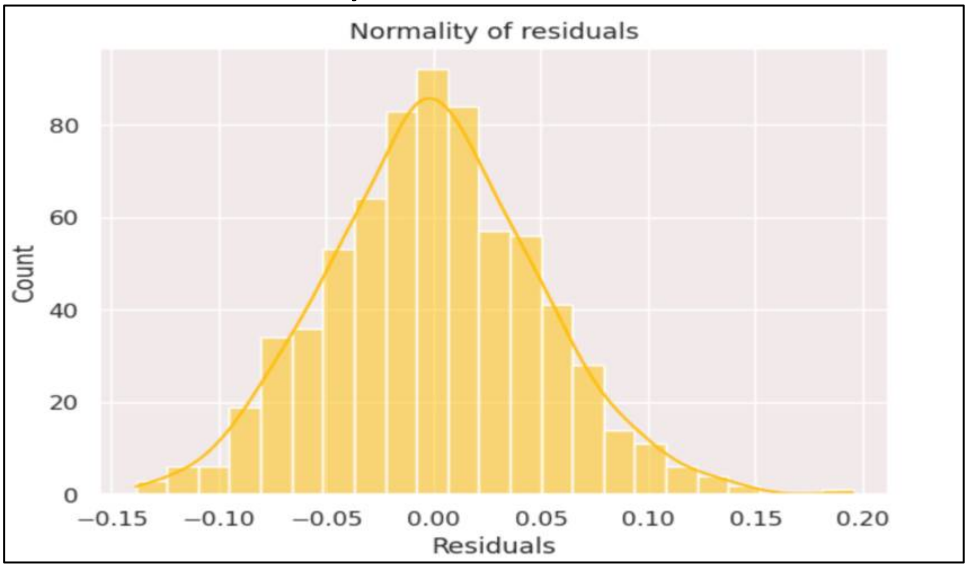

Figure 30. Histogram Plot of Residuals.

Observations

- The histogram of residuals does have a bell curve.
- Therefore, **normality of residuals assumptions is satisfied**.

Q-Q Plot of Residuals For Normality Check:

We used SciPy probplot() to plot the graph.

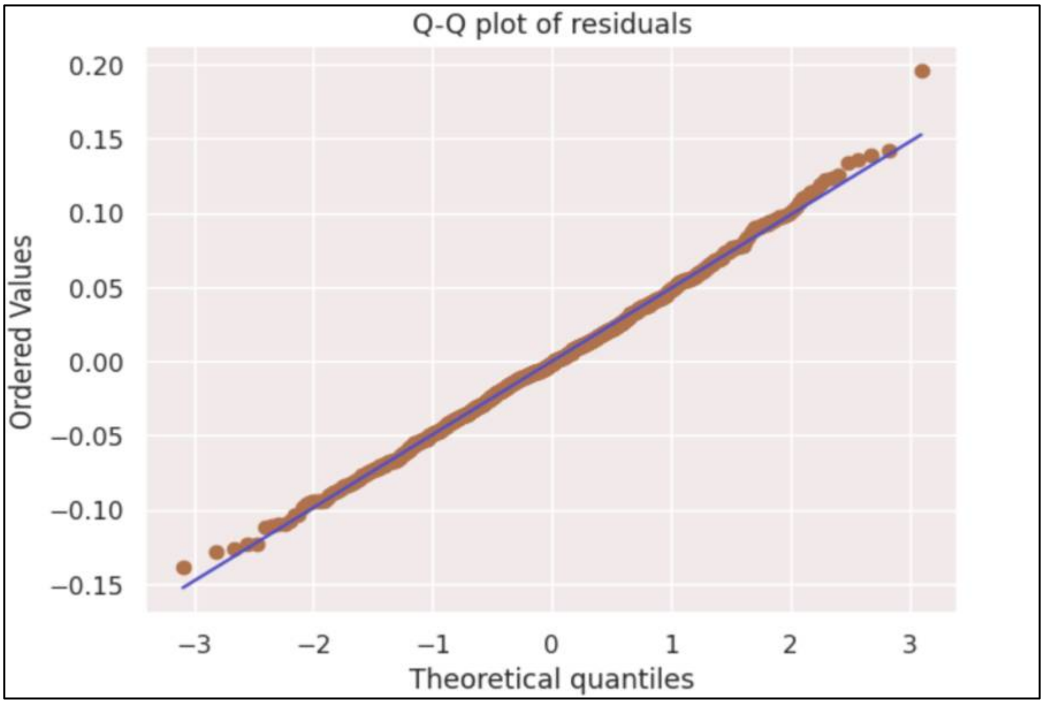

Figure 31. Q-Q Plot of Residuals.

Observations

- We can clearly see from the plot that the blue points follow the red diagonal line quite closely over the centre range.
- This indicates that the residuals are approximately normally distributed.
- Even though slight deviation at the end (tails), suggesting there are a few points that slightly deviate from the red line at both extremes (left < -2.5 and right > +2.5).
- This is normal for real-world data and **does not violate the normality assumption.**

Shapiro-Wilk's Test for Checking Residuals Normality:

Null hypothesis: The residuals are regularly distributed.
Alternative hypothesis: The residuals are not normally distributed.

Shapiro's Result – Statistic = 0.9974
P-value = 0.3457

Observations:

- Since **p-value (0.3457) > 0.05**, we **fail to reject null** hypothesis, the residuals are normal as per the Shapiro-Wilk test.
- **So, the normality assumption is satisfied.**

Therefore, we can say our residuals are normal and assumption is satisfied.

5.4 Test For Homoscedasticity

Homoscedascity: If the variance of the residuals is symmetrically distributed throughout the regression line, the data is considered homoscedastic.

Heteroscedascity: If the variance of the residuals across the regression line is unequal, the data is considered heteroscedastic.

Need for the test:

- The presence of non-constant variance in the residuals causes heteroscedasticity.
- Non-constant variance occurs in the presence of outliers.

checking for homoscedasticity:

- The **residual vs fitted values plot can be used to test for homoscedasticity**. We already plotted this above and we observed was no pattern. Thus, **as per plot residuals are homoscedasticity**
- Heteroscedasticity can result in non-symmetrical residual shapes, such as arrows. The **Goldfeld-Quandt test is utilised here using het_goldfeldquandt()**. If the p-value exceeds 0.05, we can conclude that the residuals are homoscedastic. Otherwise, they're heteroscedastic.
  - Null hypothesis (H_0): The residuals are homoscedastic.
  - Alternative hypothesis (H_1): Residuals exhibit heteroscedasticity
- If this assumption is not followed then heteroscedasticity can be reduced by incorporating more essential features or performing transformations.

F statistic = 1.18148
P-value = 0.06287

Observations

- Since p-value **(0.062) > 0.05**, **fail to reject H_0**, no evidence of heteroscedasticity.
- **We can say that the residuals are homoscedastic. So, this assumption is satisfied.**

All the assumptions of linear regression model are satisfied.

6 Final Model & PERFORMANCE CHECK

After testing all the assumptions we created the final model using linear regression.

6.1 Final Model Building

We build the model and fit the model using sm.OLS(), fit() on final trained data.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.786
Model:                            OLS   Adj. R-squared:                  0.783
Method:                 Least Squares   F-statistic:                     229.7
Date:                Wed, 09 Jul 2025   Prob (F-statistic):          8.48e-222
Time:                        10:03:26   Log-Likelihood:                 1115.3
No. Observations:                 700   AIC:                            -2207.
Df Residuals:                     688   BIC:                            -2152.
Df Model:                          11
Covariance Type:            nonrobust
=========================================================================================
                            coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                    -0.6842      0.026    -26.269      0.000      -0.735      -0.633
visitors                  0.1295      0.008     16.029      0.000       0.114       0.145
major_sports_event       -0.0633      0.004    -16.202      0.000      -0.071      -0.056
views_trailer             0.2213      0.005     42.081      0.000       0.211       0.232
dayofweek_Monday          0.0286      0.012      2.420      0.016       0.005       0.052
dayofweek_Saturday        0.0563      0.007      7.885      0.000       0.042       0.070
dayofweek_Sunday          0.0348      0.008      4.472      0.000       0.019       0.050
dayofweek_Thursday        0.0135      0.007      2.006      0.045       0.000       0.027
dayofweek_Wednesday       0.0449      0.004     10.095      0.000       0.036       0.054
season_Spring             0.0248      0.005      4.636      0.000       0.014       0.035
season_Summer             0.0447      0.005      8.310      0.000       0.034       0.055
season_Winter             0.0283      0.005      5.339      0.000       0.018       0.039
==============================================================================
Omnibus:                        4.279   Durbin-Watson:                   1.964
Prob(Omnibus):                  0.118   Jarque-Bera (JB):                4.119
Skew:                           0.162   Prob(JB):                        0.128
Kurtosis:                       3.190   Cond. No.                         66.3
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Table 16. Final Regression Model.

Final Model Coefficients:

	Feature	Coefficient
0	const	-0.684187
1	visitors	0.129517
2	major_sports_event	-0.063322
3	views_trailer	0.221317
4	dayofweek_Monday	0.028619
5	dayofweek_Saturday	0.056312
6	dayofweek_Sunday	0.034750
7	dayofweek_Thursday	0.013534
8	dayofweek_Wednesday	0.044889
9	season_Spring	0.024770
10	season_Summer	0.044741
11	season_Winter	0.028310

Table 17. Final Model Coefficients.

- Final model has only **11 features**.

6.2 Final Model Performance Check

Checking final model performance on both train and test data.

Final Model Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE (%)
Train	0.049184	0.038688	0.785965	0.782227	8.603114
Test	0.050850	0.040957	0.764165	0.754304	9.114764

Figure 32. Final Model Performance Parameters.

Observations

- The model is able to **explain ~76% of the variation in the data.**
- The **train and test RMSE and MAE are low** and similar, showing that the model generalises well with slight overfitting, which is acceptable and expected in real-world scenarios.
- The **MAPE** on the test set suggests we can predict within **9.11%** of the content views.
- Hence, **we can conclude the final model is good for prediction as well as inference purposes.**

6.3 Original & Final Model Performance Comparison

We compared the performance of initial and final model on various performance metrics.

Training performance comparison:

	Linear Regression (original)	Linear Regression (final)
RMSE	0.048844	0.049184
MAE	0.038426	0.038688
R-squared	0.788913	0.785965
Adj. R-squared	0.782374	0.782227
MAPE (%)	8.549048	8.603114

Test performance comparison:

	Linear Regression (original)	Linear Regression (final)
RMSE	0.050487	0.050850
MAE	0.040592	0.040957
R-squared	0.767519	0.764165
Adj. R-squared	0.749958	0.754304
MAPE (%)	9.009553	9.114764

Figure 33. Original vs. Final Model Performance.

Original vs. Final Model Overall Observations & Interpretations

R-squared and Adjusted R-squared:

- ***R-squared Train***
  - **Original: 0.7889**
  - **Final: 0.7860**
- ***R-squared Test***
  - **Original: 0.7675**
  - **Final: 0.7641**

- After removing high p-value predictors, **the final model's R^2 shows a little decline (≈0.002-0.003)**, but it still explains around **76% of the variation in test data**.

Adjusted R-squared:

- ***Adjusted R-squared Train***
  - **Original value: 0.7824**
  - **Final value: 0.7822**
- ***Adjusted R-squared Test***
  - **Original value: 0.7500**
  - **Final value: 0.7543 (a slight improvement)**
- The **final model's Adjusted R^2 on test data improved marginally, indicating that even though it has fewer variables, it still generalises well.**

Model coefficients (significance):

- **All predictors now have p-values < 0.05**, indicating that all variables in the model are statistically significant.
- Key variables with the biggest influence (high t-values) is **views_trailer (t ≈ 42) and it is the primary driver of viewership.**
- Other key Predictors: visitors, major_sports_event, dayofweek_Wednesday, and season_Summer all contribute significantly.

Error Metrics:

1. Mean Absolute Error (MAE)

- ***Train MAE***
  - **Original MAE: 0.0384.**
  - **Final MAE value: 0.0387.**
  - The **difference is minimal (0.0003), indicating similar average error on the observed data**.
- ***Test MAE***
  - **Originally MAE: 0.0406**
  - **Final MAE value: 0.0410**.
  - Again, a little rise (0.0004), but still within acceptable limits.
- The average error in predictions increased slightly but remained very low, approximately 40,000 views off, which is acceptable for business purposes.

2. Root Mean Squared Error (RMSE)

- ***Train RMSE***
  - **Original: 0.0488.**
  - **Final value: 0.0492**
- ***Test RMSE***
  - **Original: 0.0505**
  - **Final: 0.0508**
- **RMSE is also extremely steady. Although the final model includes fewer features, the error is still around 0.05M, showing good predictive accuracy**.

3. Mean Absolute Percentage Error (MAPE)

- ***MAPE Train***
  - **Original: 8.55%.**
  - **Final: 8.60%**
- ***MAPE Test***
  - **Original: 9.01%.**
  - **Final: 9.11%**
- The **final model predicts ±0.045M views for most content, with only a minor increase** in average percentage error (around 10%), still within the acceptable range.

Conclusion

- The **final linear regression model exhibits a great fit and consistent predictive power**:
  - **Train R^2 = 0.786, Test R^2 = 0.764,** with a slight decline (~2.2%).
  - Since, achieved an R-squared of 0.764 on the test set, explaining 76% of the variance in first-day content viewership
  - **MAPE is 8.6% (Train) vs. 9.1% (Test), while RMSE/MAE demonstrate negligible variability between training and testing sets**.
  - These tiny and acceptable variances imply that the model generalises well, which means it performs consistently on unseen data.
  - **There is no evidence of overfitting (where the model memorises the training data) or underfitting** (where the model fails to learn the patterns).
- **Thus, we conclude that the final model is more statistically solid, interpretable, and commercially viable for identifying important drivers of first-day content viewership on the OTT platform.**

6.4 PREDICTIONS ON TEST (UNSEEN) DATA

We did prediction test to check how well our model performs on unseen (test) data. Here, **Actual values are true target values (views_content) and Predicted values are of model predicted values.** We also verified the predicted values accuracy by finding mean absolute error.

	Actual	Predicted
983	0.43	0.444878
194	0.51	0.518432
314	0.48	0.437527
429	0.41	0.492618
267	0.41	0.484167
746	0.68	0.644531
186	0.62	0.619178
964	0.48	0.502187
676	0.42	0.491791
320	0.58	0.591065

Table 18. Predictions on Test Data.

Average Absolute Error: 0.04095

Observations

- **The final model's average absolute error on the test set is around 0.041 million views, implying that forecasts differ from actual first-day views by around 41,000 views on average.**
- This **inaccuracy is rather small**, especially given that the target values (views_content) range around between 0.30 and 0.80 million.
- Given that the **Mean Absolute Percentage Error (MAPE) was also approximately 9%, this level of error is reasonable for business decision-making and indicates that the model makes solid predictions on unseen data.**

7 ACTIONABLE INSIGHTS

Model Performance

- The final linear regression model is highly predictive, with a **test R-squared of 0.764 (train = 0.786)**, explaining **76.4%** of the variance in first-day content viewership.
- The model exhibits strong generalisation with minimal overfitting (only 2.2% drop from training to test) and retains commercial viability with a **MAPE of 9.11%**, suggesting predictions tend to be within ±9% of actual viewership.

Significance of Predictors

Primary Drivers

1. views_trailer(Coefficient: 0.221, 95% CI: [0.211, 0.232])

- The **most significant predictor** with the highest statistical influence (**$p < 0.001$, $t = 42.081$**).
- Since, the coefficient (**0.221**) is positive, it has positive linear relationship with the target variable.
- There is a **0.221 million** increase in content views for every unit increase in trailer views.
- The true effect is between **0.211 and 0.232 million views**, as per the confidence interval.
- Therefore, trailer views is the best indicator of first-day viewership success.

2. visitors (Coefficient: 0.130; 95% confidence interval: [0.114, 0.145])

- The second most significant predictor (**$t = 16.029$, $p < 0.001$.**).
- The coefficient (**0.130**) is positive, indicating a positive linear relationship with the target variable.
- Each unit increase in visitors increases content views by **0.130 million views**.
- The confidence interval confirms a positive effect between **0.114 and 0.145 million views**.
- This shows a direct relationship between platform traffic and content success.

3. major_sports_event (Coefficient: -0.063; 95% CI: [-0.071, -0.056])

- A significant negative impact on content viewership (**$t = -16.202$, $p < 0.001$**).
- Since, the coefficient (**-0.063**) is negative, it suggests a negative linear relationship with the target variable.
- Sports events decrease content views by **0.063 million views** with high confidence.
- The confidence interval reveals a consistent negative effect of -0.071 to -0.056 million views.
- This is a significant drop, suggesting that sports events considerably reduce viewership.

Temporal Patterns

1. Day of Week Effects:

- Saturday (**coef: 0.056, 95% CI: [0.042, 0.070]**) shows the strongest positive impact on content views (**$t = 7.885$, $p < 0.001$**), indication weekend drive higher viewership. Thus, launching content on Saturdays will be beneficial, as it is likely to increase viewership.
- Wednesday (**coef: 0.045, 95% CI: [0.036, 0.054]**) displays second highest weekday influence, with consistently strong effect (**$t = 10.095$, $p < 0.001$**). It shows the strongest statistical significance, suggesting a highly reliable effect on boosting views mid-week.
- Sunday (**coef: 0.035, 95% CI: [0.019, 0.050]**) offers a moderate weekend advantage in views(**$t = 4.472$, $p < 0.001$**). It is a favourable day for content drops.
- Monday (**coef: 0.029, 95% CI: [0.005, 0.052]**) has smaller but still statistically significant increase in views(**$t = 2.420$, $p = 0.016$**). This indicates Mondays can support light content launches.
- Thursday (**coef: 0.014, 95% CI: [0.000, 0.027]**) has a lowest positive impact, but remains statistically significant (**$t = 2.006$, $p = 0.045$**). The day is suitable for experimental or niche content releases.

2. Seasonal Variations:

Summer

- Summer shows the **highest positive effect** on first-day content views.
- Coefficient: **0.045, 95% CI: [0.034, 0.055], $t = 8.310$, $p < 0.001$**

- Thus, Summer brings the strongest seasonal boost in viewership.

Winter

- Winter has a **moderate yet significant effect**.
- Coefficient: **0.028, 95% CI: [0.018, 0.039], t = 5.339, p < 0.001**
- Winter contributes a meaningful uplift to content views, second only to summer.

Spring

- Spring provides the **smallest but still significant impact**.
- Coefficient: **0.025, 95% CI: [0.014, 0.035], t = 4.636, p < 0.001**
- Spring has a lesser influence, yet its effect is statistically reliable.

Main Inferences

- All predictors in the final model are statistically significant ($p < 0.05$) and have narrow confidence intervals that do not include zero, enhancing the reliability of predictor estimates.
- The model's commercial relevance for business decision-making is validated by its consistent performance across training and testing datasets **(RMSE: 0.0508, MAE: 0.0410)**.

Model Limitations

- In order to improve prediction accuracy, future research could investigate interaction effects, such as how trailer views vary across seasons or sporting events, which are not taken into consideration by the current model.

8 BUSINESS RECOMMENDATIONS

1. Marketing Strategy

Prioritise Trailer Investment:

- Channel digital ad budget is based on trailer-to-view ratio criteria. Give top priority to high-ROI trailer forms, like short-form and platform-native trailers.
- Set trailer view targets as the primary indicators of content performance.

Improve Web Traffic Generation:

- Utilise internal analytics to evaluate landing pages for visitor-to-view conversion.
- Channel marketing spends on audience building and conversion optimisation. Develop a content marketing plan to increase organic traffic.

2. Scheduling Optimisation

Strategic Release Timing:

- Schedule premium content releases on Wednesdays and Saturdays to maximise impact.

- Use Sunday and Monday for secondary content launches.
- Avoid significant releases on baseline days (Tuesday and Friday).

Seasonal Content Planning:

- Launch significant campaigns over the summer for maximum seasonal increase.
- Plan secondary releases throughout the winter months.
- Use spring for audience growth and content preparation.

3. Competitive Strategy

Mitigating Sports Events:

- Create an automated sports event calendar to prevent scheduling conflicts.
- Create a counter-programming approach for unavoidable competitive instances.
- Implement content release blackout periods during major sporting events.

4. Performance Monitoring

Model-Based Decision Making:

- Implement real-time monitoring of all crucial predictors.
- Use the model's 76.4% explanatory power to predict content performance.
- Set up automated warnings for performance deviations from forecast values.

5. Estimated Business Impact

- **Immediate impact (3-6 months):** Improved accuracy in estimating first-day viewing by 15-20% through data-driven scheduling and trailer optimisation.
- **Strategic shift (6-12 months):** Season-aware and trailer-first strategy alignment leads to a ~25-30% boost in platform-wide engagement.
- **Long-term transformation (over 12 months):** Implementation of predictive analytics for release planning and marketing ROI optimisation.
- The model's statistical robustness **(R^2 = 0.764, MAPE = 9.11%)** indicates that these recommendations will lead to **significant improvements in OTT platform content performance**.